CHAPTER 14
LIGHT

3.2 Light

3.2.1 Reflection of light

1 Define and use the terms normal, angle of incidence and angle of reflection

2 Describe an experiment to illustrate the law of reflection

3 Describe an experiment to find the position and characteristics of an optical image formed by a plane mirror (same size, same distance from mirror as object and virtual)

4 State that for reflection, the angle of incidence is equal to the angle of reflection and use this in constructions, measurements and calculations

3.2.2 Refraction of light

1 Define and use the terms normal, angle of incidence and angle of refraction

2 Define refractive index n as $n = \frac{\sin i}{\sin r}$; recall and use this equation

3 Describe an experiment to show refraction of light by transparent blocks of different shapes

4 Define the terms critical angle and total internal reflection; recall and use the equation

$$n = \frac{1}{\sin c}$$

5 Describe experiments to show internal reflection and total internal reflection

6 Describe the use of optical fibres, particularly in telecommunications, stating the advantages of their use in each context

3.2.3 Thin lenses

1 Describe the action of thin converging and thin diverging lenses on a parallel beam of light

2 Define and use the terms focal length, principal axis and principal focus (focal point)

3 Draw ray diagrams to illustrate the formation of real and virtual images of an object by a converging lens and know that a real image is formed by converging rays and a virtual image is formed by diverging rays

4 Define linear magnification as the ratio of image length to object length; recall and use the equation

$$\text{linear magnification} = \frac{\text{image length}}{\text{object length}}$$

5 Describe the use of a single lens as a magnifying glass

6 Draw ray diagrams to show the formation of images in the normal eye, a short-sighted eye and a long-sighted eye

7 Describe the use of converging and diverging lenses to correct long-sightedness and short-sightedness

3.2.4 Dispersion of light

1 Describe the dispersion of light as illustrated by the refraction of white light by a glass prism

2 Know the traditional seven colours of the visible spectrum in order of frequency and in order of wavelength

1. O/N/23/11/Q18

A plane mirror on a vertical wall forms an image of an object placed in front of it.

Which characteristics describe the image?

A real and inverted

B real and upright

C virtual and inverted

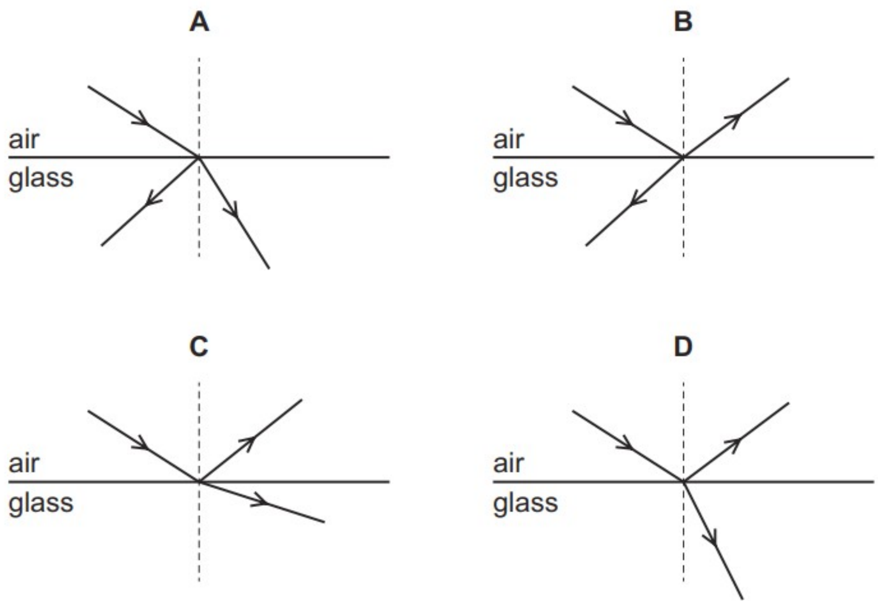
D virtual and upright

2. O/N/23/11/Q19

The diagrams show light travelling in air incident on the surface of a glass block.

Some light is reflected and some light is refracted.

Which diagram shows the reflection and refraction of the light?



3. O/N/23/11/20

Which statement is correct?

A Total internal reflection only occurs when light travels from air into glass.

B The larger the refractive index of glass, the larger is the critical angle.

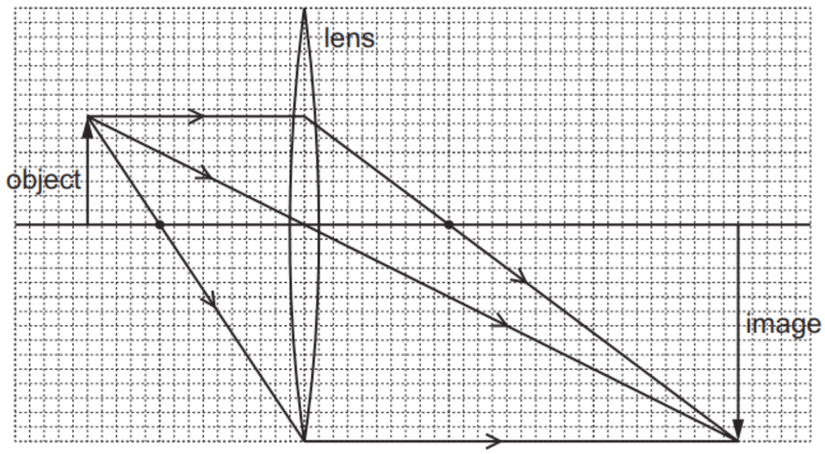
C When total internal reflection occurs, the angle of incidence is equal to the angle of reflection.

D When total internal reflection occurs, the angle of incidence is less than the critical angle.

4. O/N/23/11/Q21

An object is placed in front of a converging lens of focal length 4.0 cm. The height of the image is 6.0 cm.

The arrangement is shown on the scale diagram.



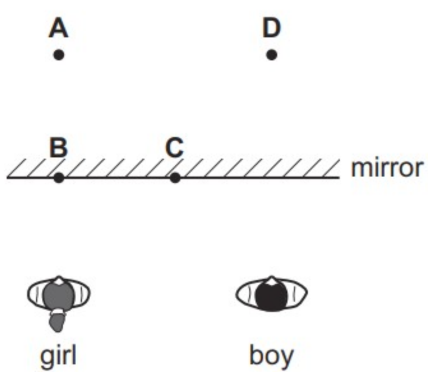
What is the linear magnification produced by the lens?

A 0.50 **B** 1.5 **C** 2.0 **D** 6.0

5. O/N/23/12/Q26

A boy stands beside a girl in front of a large vertical plane mirror. They are equal distances from the mirror, as shown. The boy sees an image of the girl.

Where is the girl's image?



6. O/N/23/12/Q27

Light travelling in air is incident on a water surface.

The refractive index of water is 1.3 and the angle of refraction in the water is 40°.

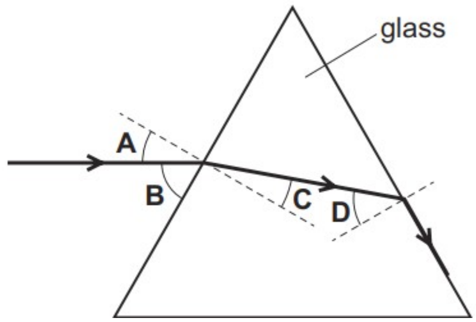
What is the angle of incidence in the air?

A 30° **B** 50° **C** 52° **D** 57°

7. O/N/23/12/Q28

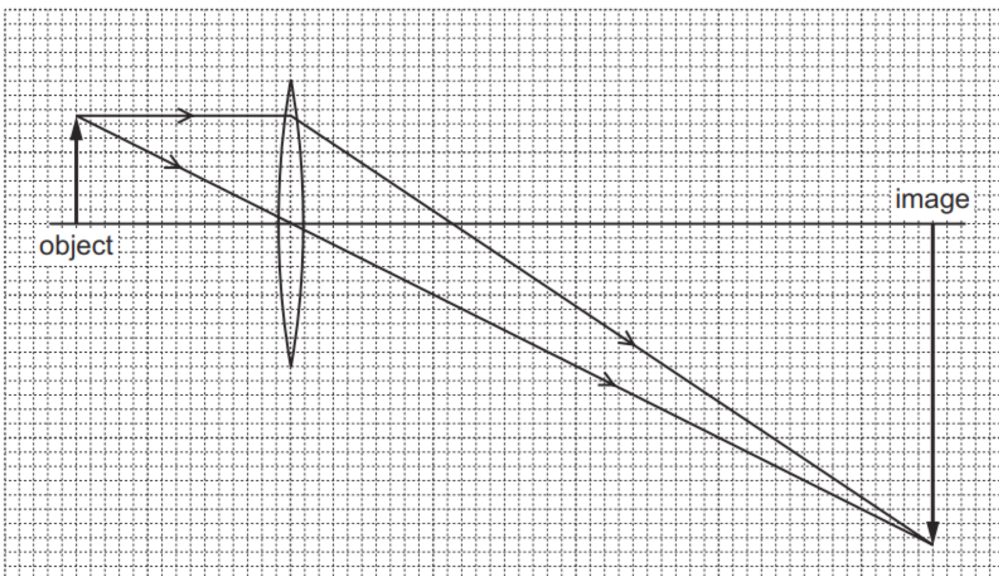
The diagram shows a ray of light passing through a triangular glass block.

Which angle is the critical angle of light in glass?



8. O/N/23/12/Q29

A lens is used to produce a magnified image, as shown in the scale diagram.

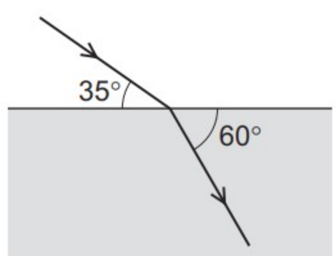


What is the linear magnification produced by the lens?

A 0.33 **B** 3.0 **C** 4.0 **D** 6.0

9. M/J/23/12/Q20

A ray of light passes into a block of transparent material as shown.



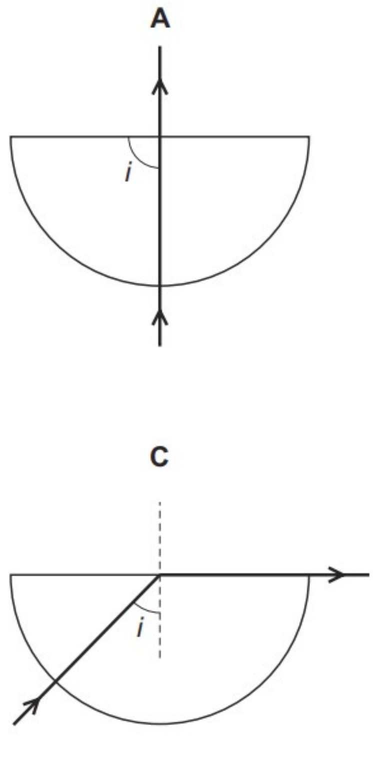
What is the refractive index of the transparent material?

A 0.66 **B** 1.15 **C** 1.64 **D** 1.83

10. M/J/23/12/Q21

The diagrams show light incident on the straight edge of a semi-circular glass block after passing through it.

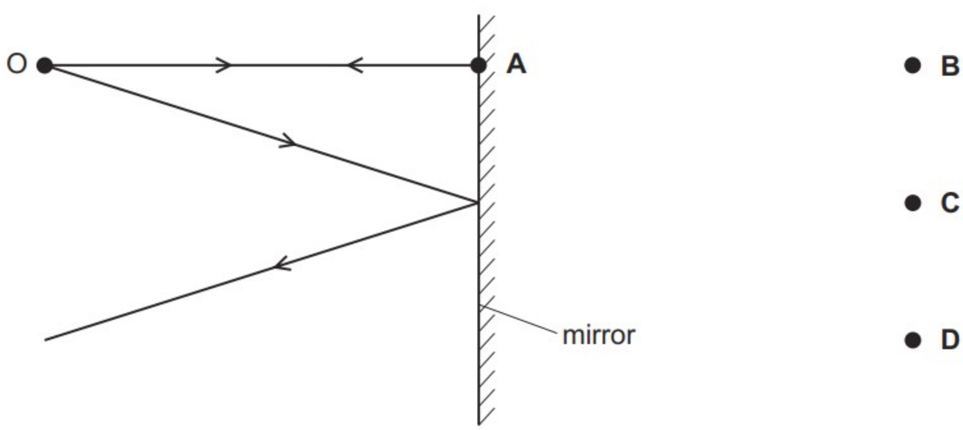
Which diagram shows refraction and an angle *i* equal to the critical angle of the glass?



11. M/J/23/11/Q19

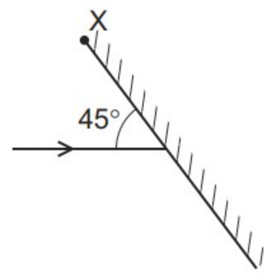
The diagram shows two divergent rays of light from an object O being reflected from a plane mirror.

At which position is the image formed?



12. M/J/23/11/Q20

Light is incident on a mirror at an angle of 45° as shown. The mirror can be rotated about an axis into the page through point X.



The mirror is rotated until the light is reflected back along its original path.

Through which angle is the mirror rotated?

A 22.5° clockwise

B 22.5° anticlockwise

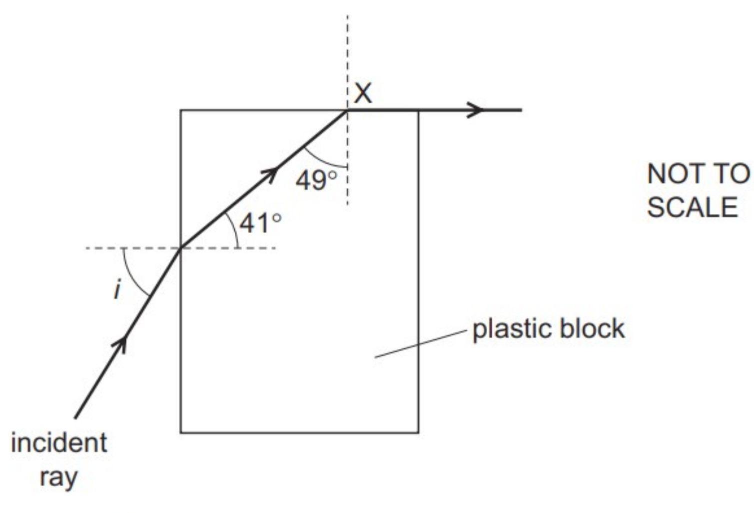
C 45° clockwise

D 45° anticlockwise

13. M/J/23/11/Q21

A ray of light is incident on a plastic block in air, at an angle of incidence *i*. The refractive index of the plastic is *n*.

The light ray refracts along the plastic-air boundary when it reaches X.



Which equation is correct?

A $\sin i = \dfrac{\sin 41°}{n}$

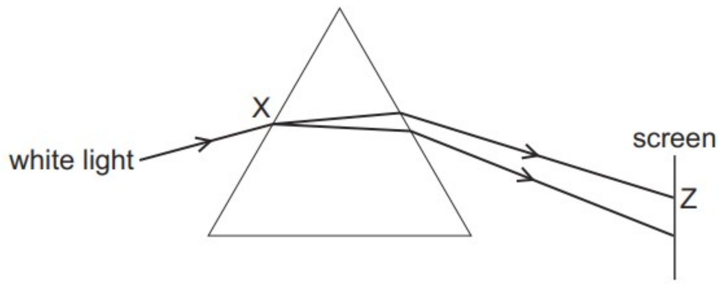
B $\sin i = \dfrac{n}{\sin 49°}$

C $\sin i = n \times \sin 41°$

D $\sin i = n \times \sin 49°$

14. M/J/23/11/Q22

The diagram shows white light entering a prism.

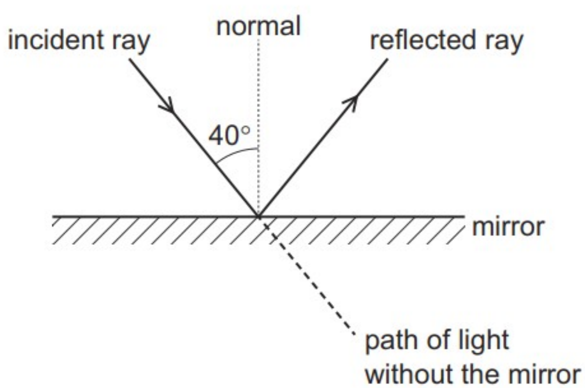


What happens at point X and what is the colour of the light that strikes the screen at point Z?

	at point X	at point Z
A	dispersion only	blue light
B	dispersion and refraction	blue light
C	dispersion and refraction	red light
D	refraction only	red light

15. O/N/22/11/Q24

A mirror is placed in the path of a ray of light.

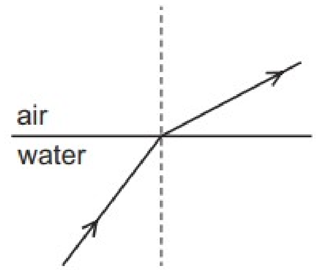


Through which angle does the direction of the ray of light change?

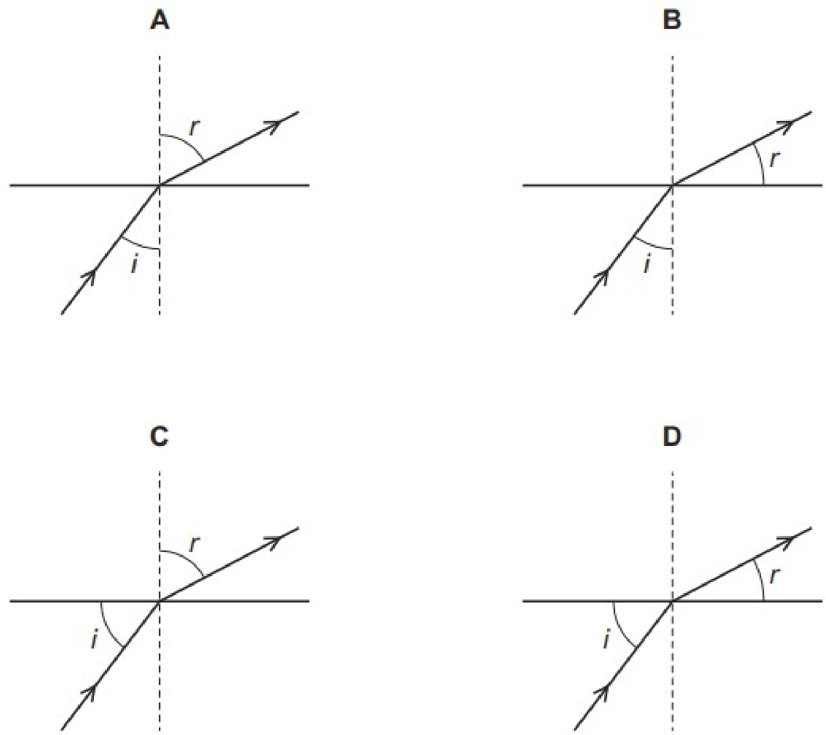
A 40° **B** 90° **C** 100° **D** 140°

16. O/N/22/12/Q23

A ray of light in water is refracted at the surface into air.

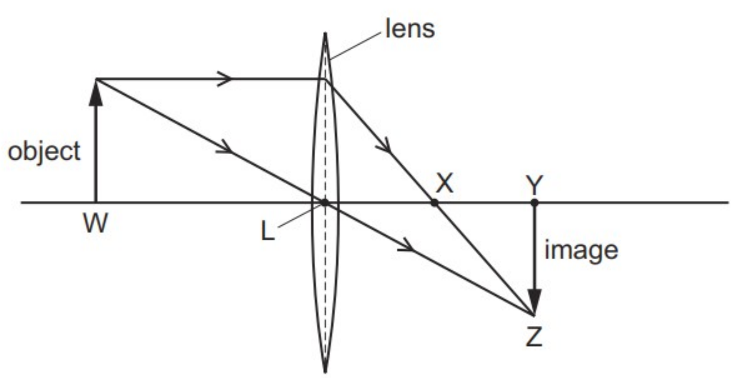


Which diagram shows the angle of incidence i and the angle of refraction r?



17. O/N/22/12/Q24

A thin converging lens forms a real, focused image of an object, as shown.

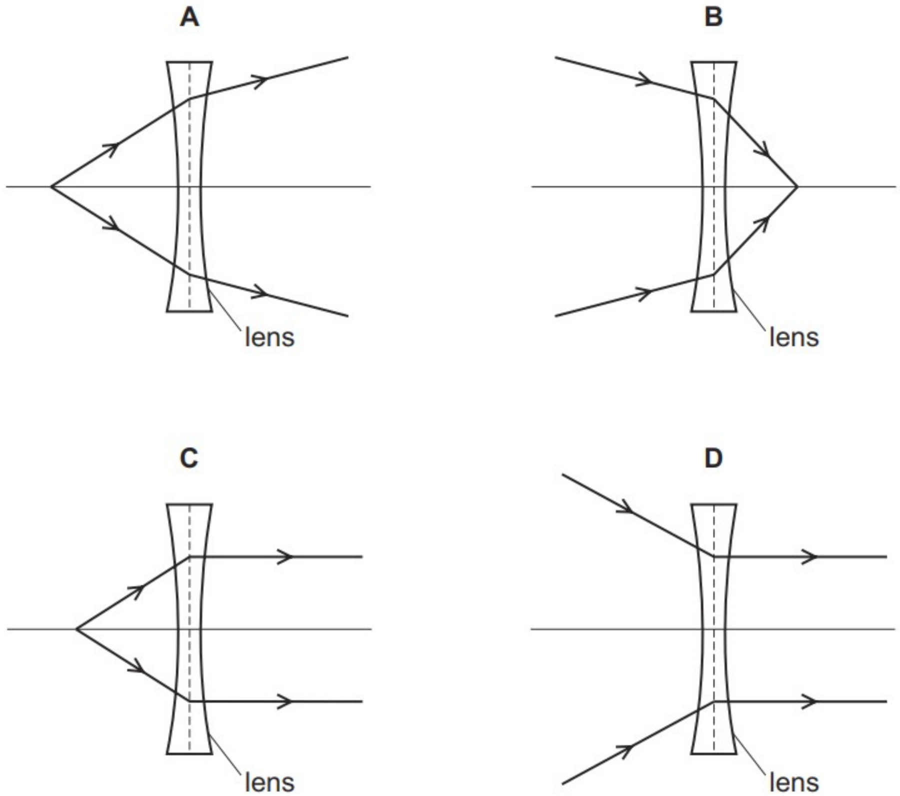


Which distance is equal to the focal length of the lens?

A LW **B** LX **C** LY **D** LZ

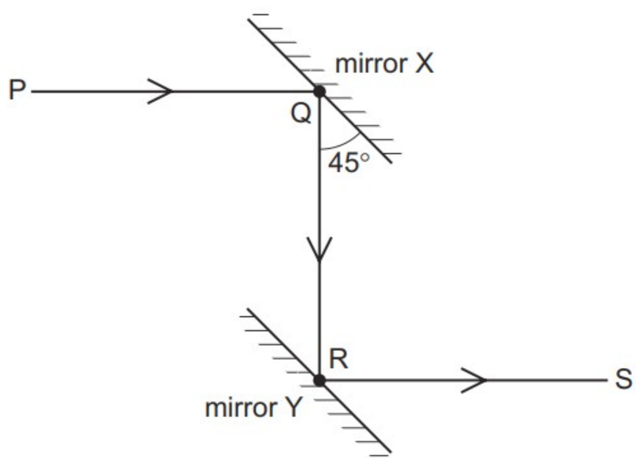
18. O/N/22/12/Q25

Which ray diagram shows the action of a diverging lens?



19. M/J/22/12/Q24

The diagram shows a ray PQ reflected by mirror X to a parallel mirror Y. The reflected ray along RS is parallel to PQ.



Which statement is correct?

A The angle between PQ and QR is 45°.

B The angle between QR and RS is 180°.

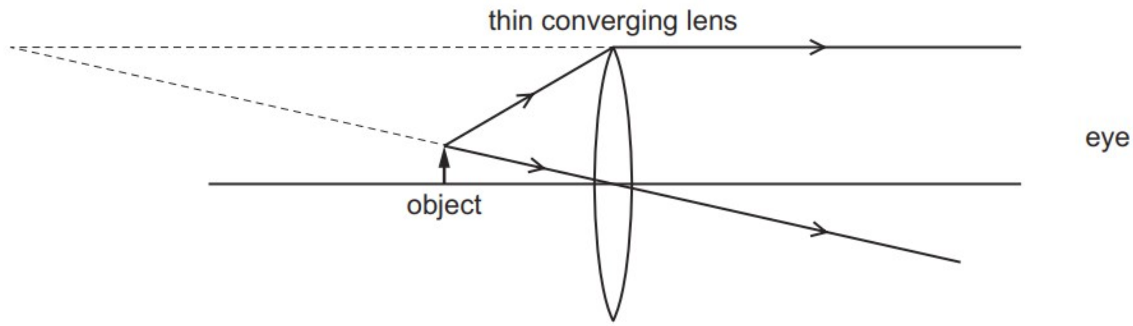
C The angle of incidence of PQ on mirror X is 60°.

D The angle of incidence of QR on mirror Y is 45°.

20. M/J/22/12/Q25

An object is viewed through a thin converging lens.

The diagram shows the paths of two rays from the top of the object to an eye.



How does the image compare with the object?

A It is larger and inverted.

B It is larger and upright.

C It is smaller and inverted.

D It is smaller and upright.

21. M/J/22/12/Q26

Violet and indigo light have the shortest wavelengths in the spectrum of visible light.

Which three colours, in order of increasing wavelength, immediately follow indigo?

A blue $\rightarrow$ green $\rightarrow$ orange

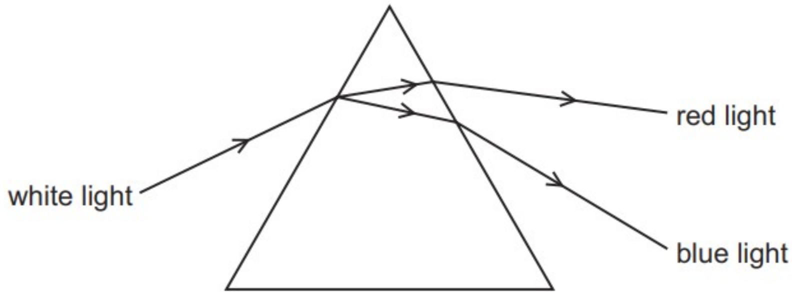
B blue $\rightarrow$ green $\rightarrow$ yellow

C green $\rightarrow$ blue $\rightarrow$ yellow

D yellow $\rightarrow$ green $\rightarrow$ orange

22. M/J/22/11/Q22

A ray of white light enters a prism as shown.



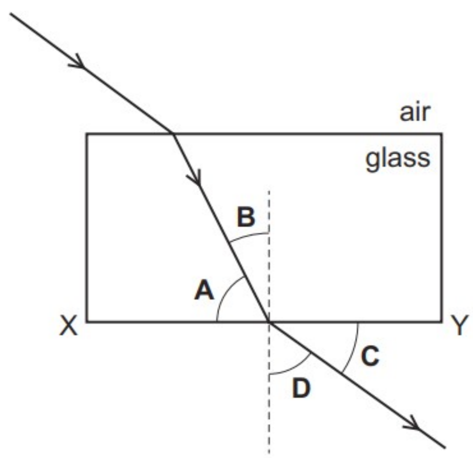
Which row is correct?

	wave properties observed	frequency of red light compared with frequency of blue light
A	dispersion only	smaller
B	refraction only	greater
C	dispersion and refraction	smaller
D	dispersion and refraction	greater

23. O/N/21/11/Q20

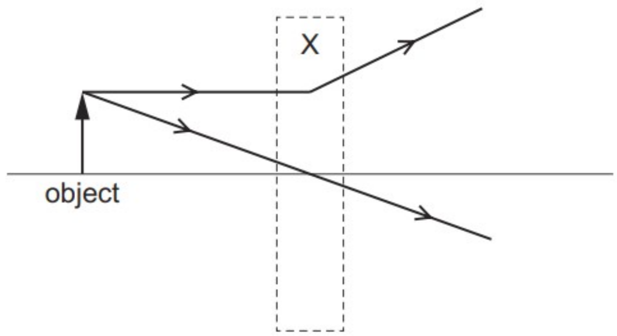
A ray of light passes into a glass block. It travels through the glass block and then emerges into the air.

Which angle is the angle of refraction at the surface XY?



24. O/N/21/11/Q21

Two rays of light pass through a lens in region X.



Which type of lens is in region X and which type of image is formed?

	type of lens	type of image
A	converging	real
B	converging	virtual
C	diverging	real
D	diverging	virtual

25. O/N/21/11/Q22

A girl is long-sighted.

Which statement is correct?

A She sees close objects less clearly than a person with normal vision.

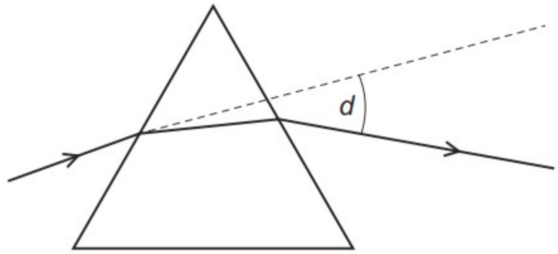
B She sees distant objects more clearly than a person with normal vision.

C The fault is corrected with a diverging lens.

D The image of a close object is formed in front of her retina.

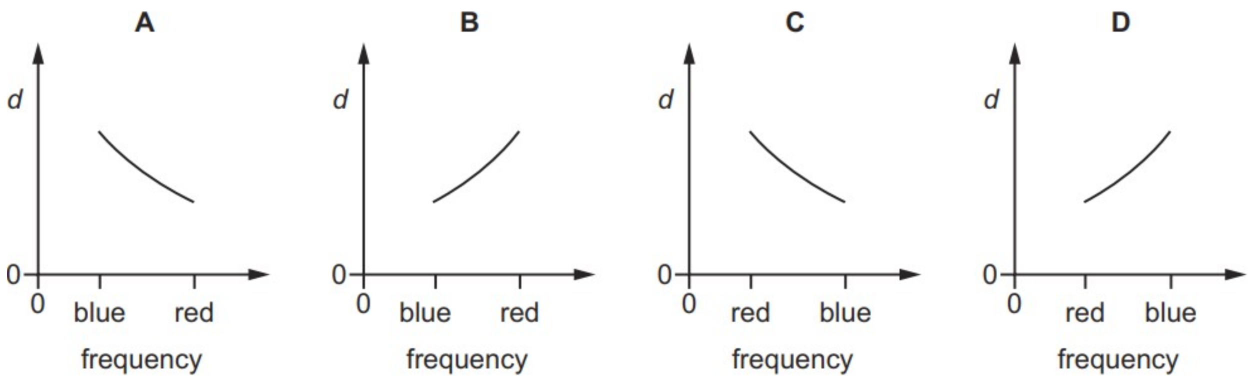
26. O/N/21/11/Q23

Light rays are deviated by a prism.



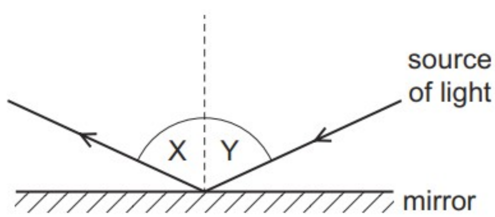
The deviation angle *d* is measured for light rays of different frequency, including blue light and red light.

Which graph of *d* against frequency is correct?



27. O/N/21/12/Q23

Light reflects from a plane mirror as shown.

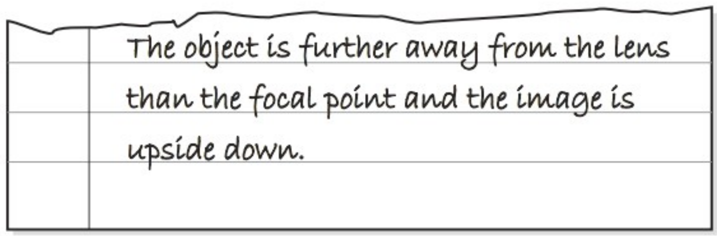


Which row is always correct?

	symbol of angle X	relationship between X and Y
A	*i*	X = Y
B	*i*	X + Y = 90°
C	*r*	X = Y
D	*r*	X + Y = 90°

28. O/N/21/12/Q24

A piece of paper torn from an exercise book is shown.



Which process is being described?

A the formation of a virtual image by a diverging lens

B the formation of a virtual image by a converging lens

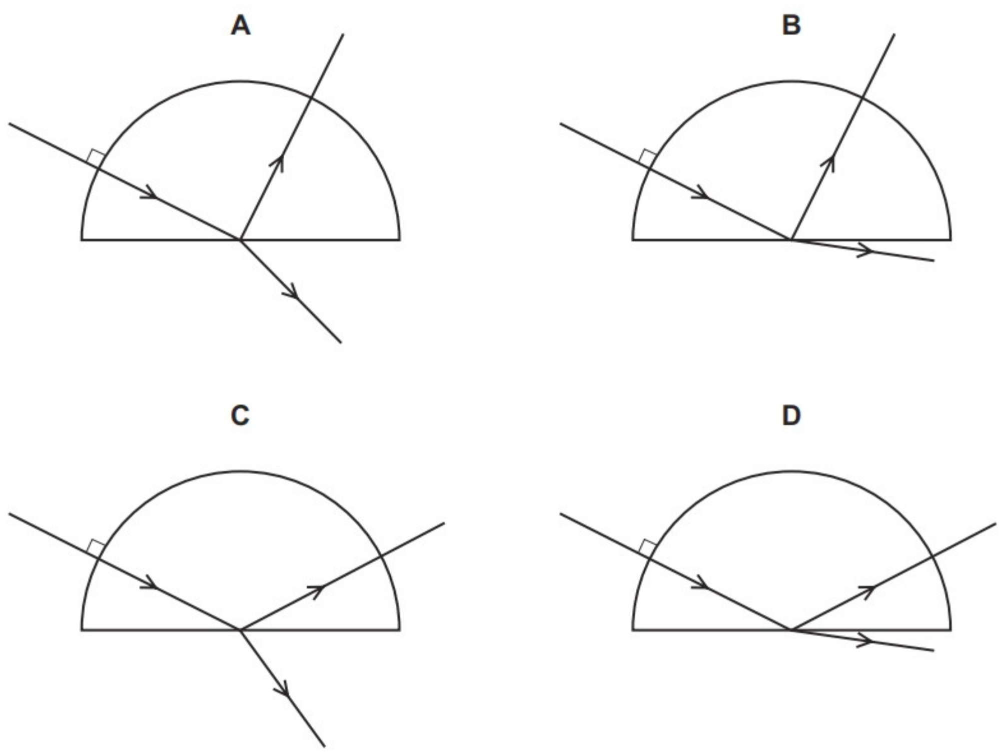
C the formation of a real image by a diverging lens

D the formation of a real image by a converging lens

29. M/J/21/12/Q24

A ray of red light in air enters a semi-circular block.

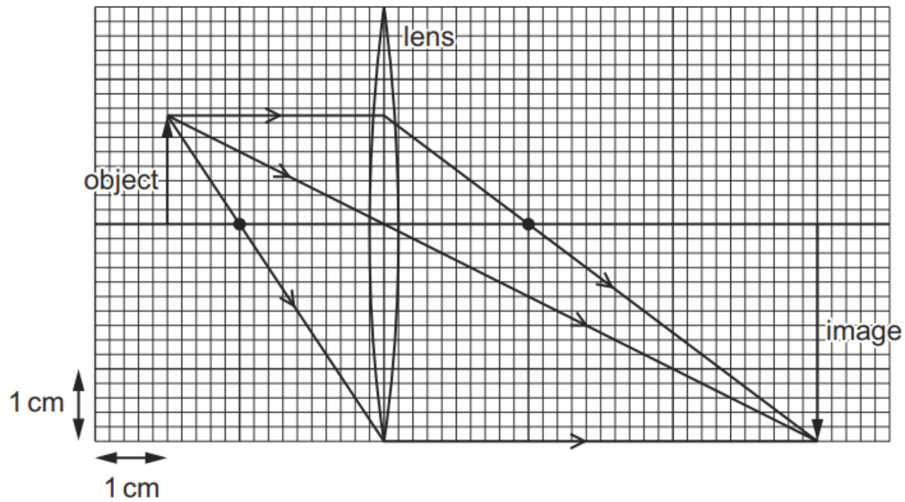
Which diagram shows the partial reflection and the refraction of the ray?



30. M/J/21/12/Q25

An object of height 1.5 cm is placed in front of a converging lens of focal length 2.0 cm.

The arrangement is shown on the full-scale ray diagram.



What is the linear magnification produced by the lens?

A 2.0 **B** 3.0 **C** 4.0 **D** 6.0

31. M/J/21/12/Q26

In which optical instrument is the distance between the object and the lens less than the focal length of the lens?

A camera

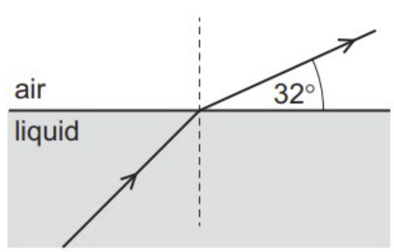
B magnifying glass

C photographic enlarger

D projector

32. M/J/21/11/Q21

Light refracts from a liquid into air as shown.



not to scale

The refractive index for light moving from air to the liquid is 1.4.

What is the angle of incidence in the liquid?

A 22° **B** 37° **C** 41° **D** 45°

33. M/J/21/11/Q23

Which statement about human vision is correct?

A In a normal eye, the image on the retina is magnified and upright.

B In a long-sighted eye, distant objects form images in front of the retina.

C Short-sighted eyes produce only virtual images.

D Short-sight is corrected by the use of a diverging lens.

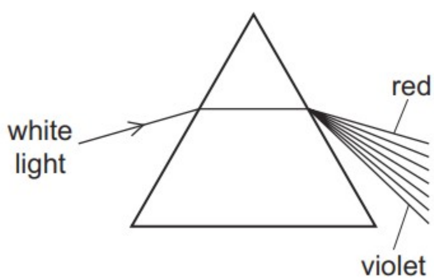
34. M/J/21/11/Q24

White light enters a prism and forms a spectrum.

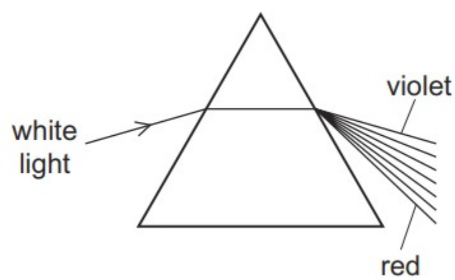
The rays in the air are labelled.

Which diagram shows how the white light is dispersed by the prism?

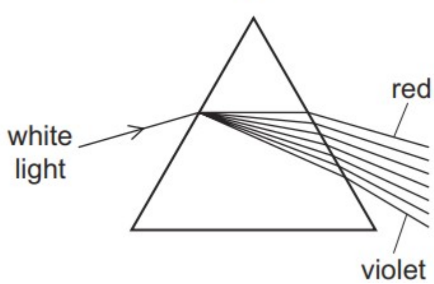
A



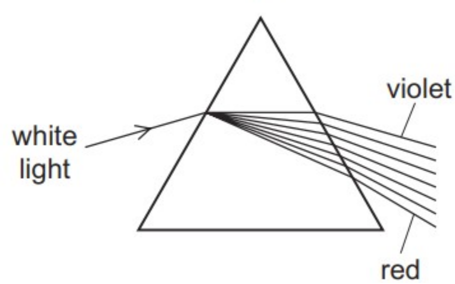
B



C

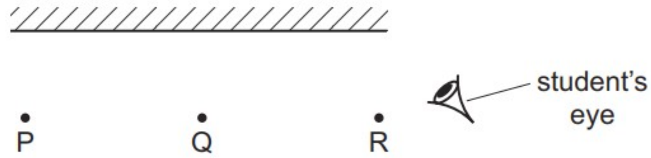


D



35. O/N/20/11/Q27

Three objects P, Q and R are placed in front of a plane mirror.



The student's eye is positioned as shown.

Which of the images of P, Q and R can the student see in the mirror?

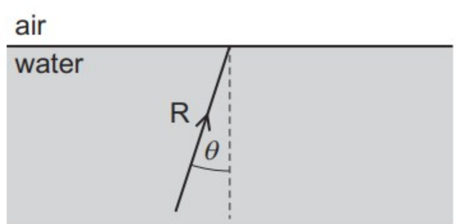
	P	Q	R
A	✓	✓	✓
B	✓	✓	✗
C	✓	✗	✗
D	✗	✗	✗

key
✓ = can see
✗ = cannot see

36. O/N/20/11/Q28

A ray of light R is incident on a water-to-air surface with an angle of incidence θ.

The angle θ is less than the critical angle *c*.



Which statement describes the subsequent path of R?

A It travels back into the water with an angle of reflection equal to *c*.

B It travels back into the water with an angle of reflection greater than *c*.

C It travels into air with an angle of refraction greater than θ.

D It travels into air with an angle of refraction smaller than θ.

37. O/N/20/12/Q24

Which statement about light passing from air to glass is correct?

A The frequency of the light waves decreases.

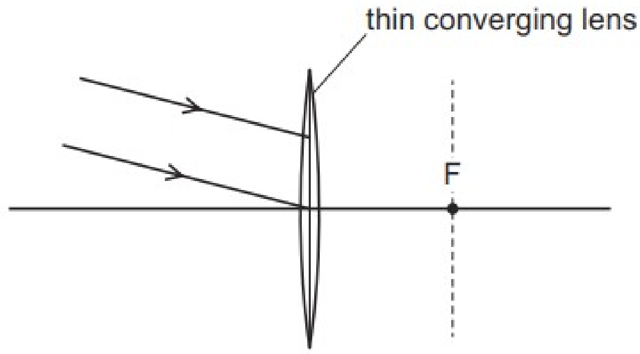
B The speed of the light waves decreases.

C The wavelength of the light waves increases.

D The wavelength of the light waves remains unchanged.

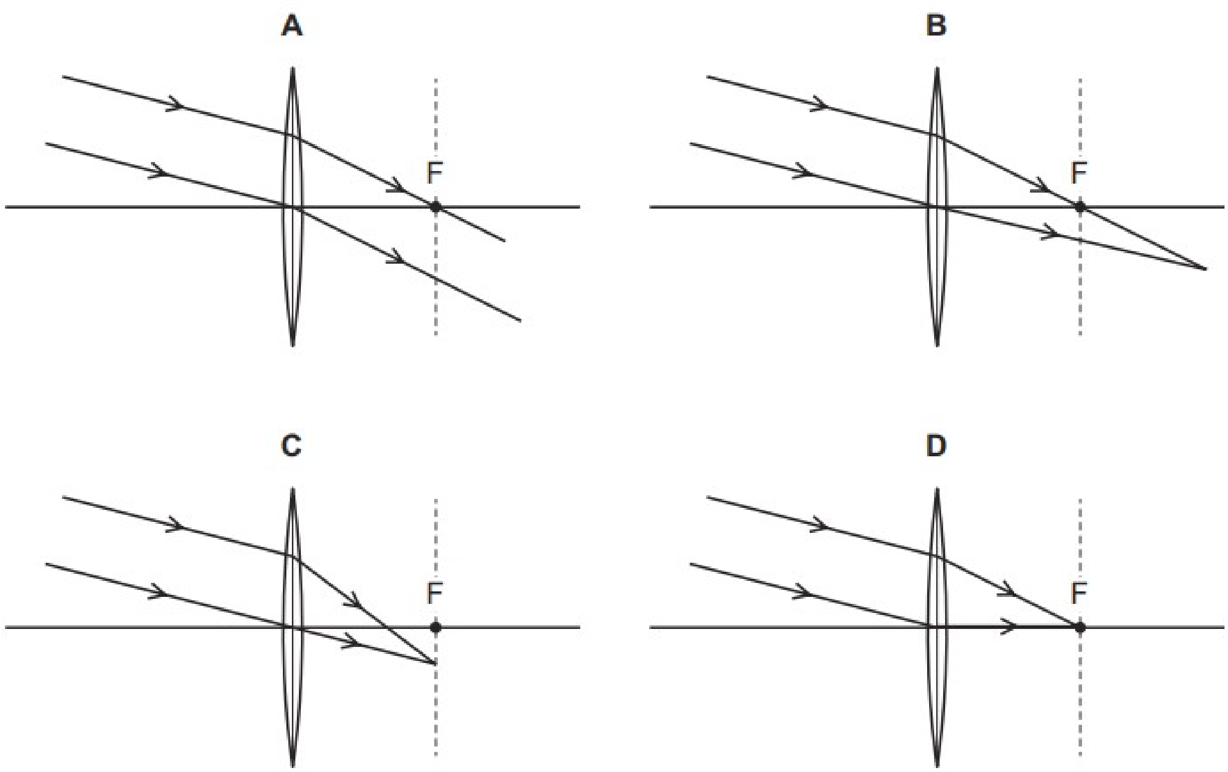
38. O/N/20/11/Q29

A parallel beam of light is incident on a thin converging lens.



F is one focal point of the lens.

Which ray diagram shows the light after it has passed through the lens?



39. O/N/20/12/Q23

A plane mirror on a vertical wall forms an image of an object placed in front of it.

Which characteristics describe the image?

A real, inverted and smaller than the object

B real, upright and the same size as the object

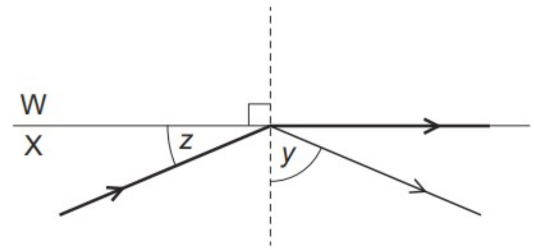
C virtual, upright and smaller than the object

D virtual, upright and the same size as the object

40. M/J/20/12/Q31

The diagram shows a ray of light incident on the boundary between two mediums W and X.

The mediums have different refractive indexes.



Some light is reflected and some passes along the surface between the two mediums. Angle *y* is greater than angle *z*.

Which statement is correct?

A W has a greater refractive index than X and angle *y* is equal to the critical angle.

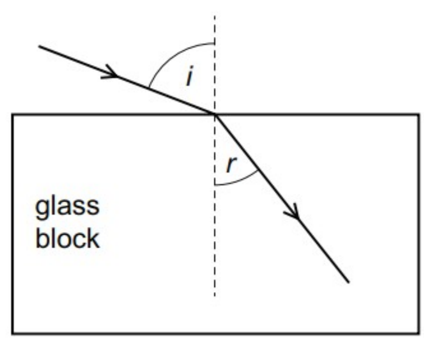
B W has a greater refractive index than X and angle *z* is equal to the critical angle.

C X has a greater refractive index than W and angle *y* is equal to the critical angle.

D X has a greater refractive index than W and angle *z* is equal to the critical angle.

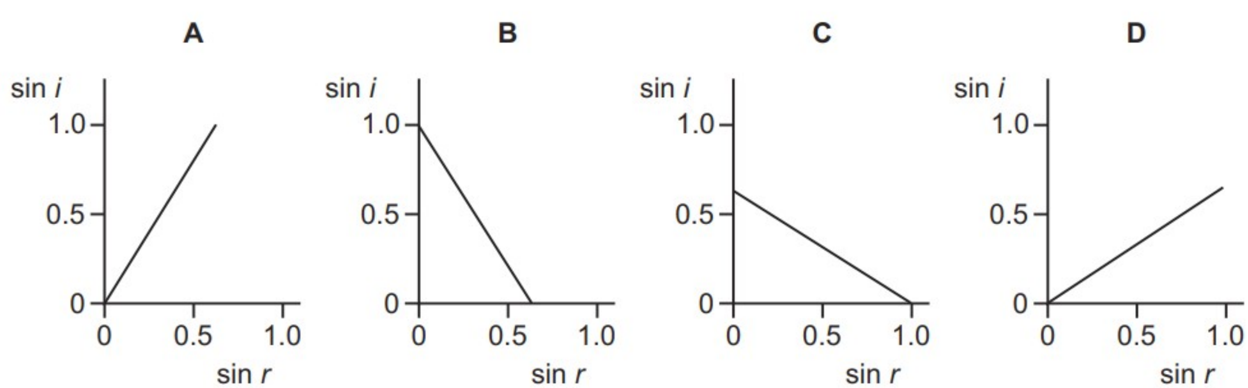
41. M/J/20/12/Q32

Light enters a glass block at an angle of incidence *i* and it produces an angle of refraction *r* in the glass.



Several different values of *i* and *r* are measured, and a graph is drawn of sin *i* against sin *r*.

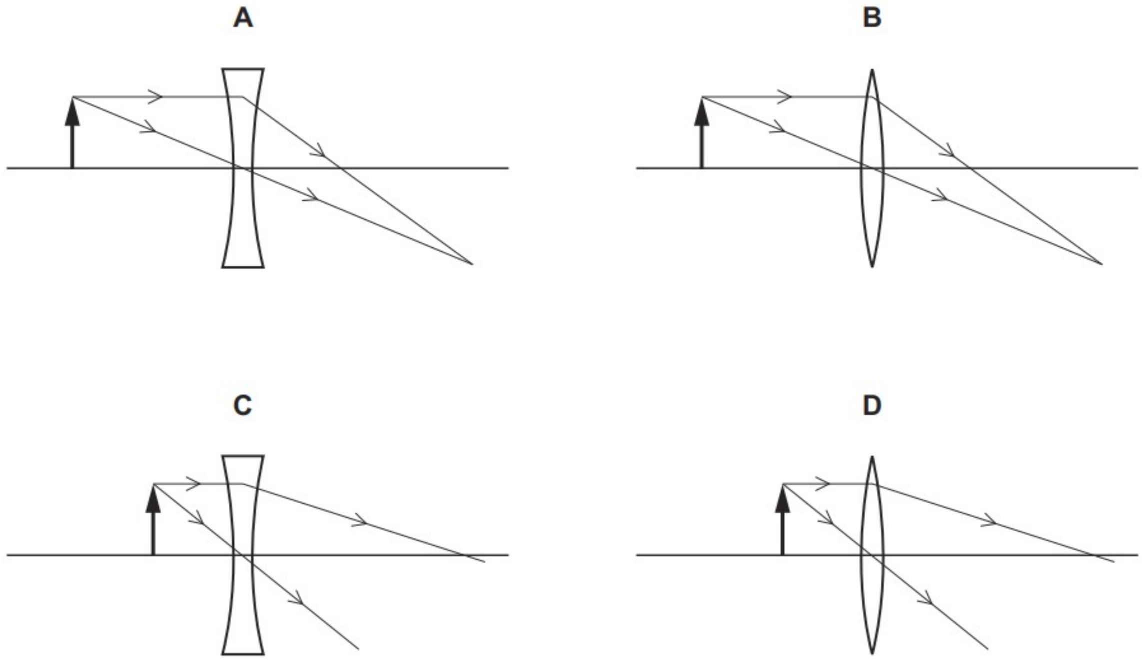
Which graph is correct?



42. M/J/20/12/Q33

A converging glass lens is used to produce a virtual, magnified image.

Which ray diagram shows the rays passing through the converging lens?

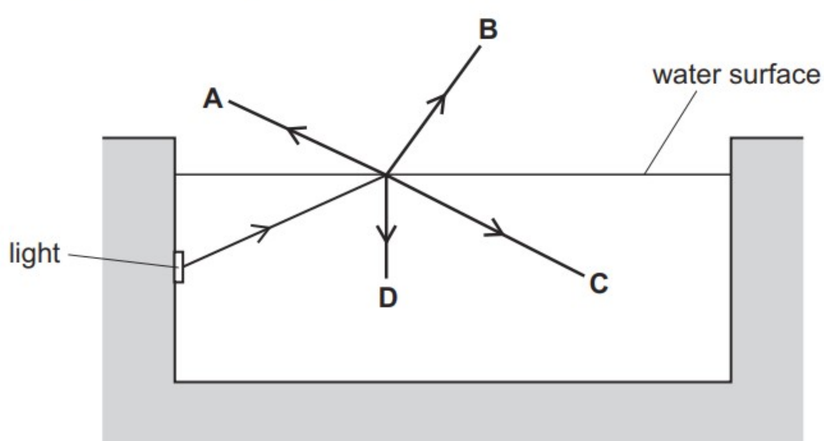


43. M/J/20/11/Q27

A swimming pool is lit by an underwater light.

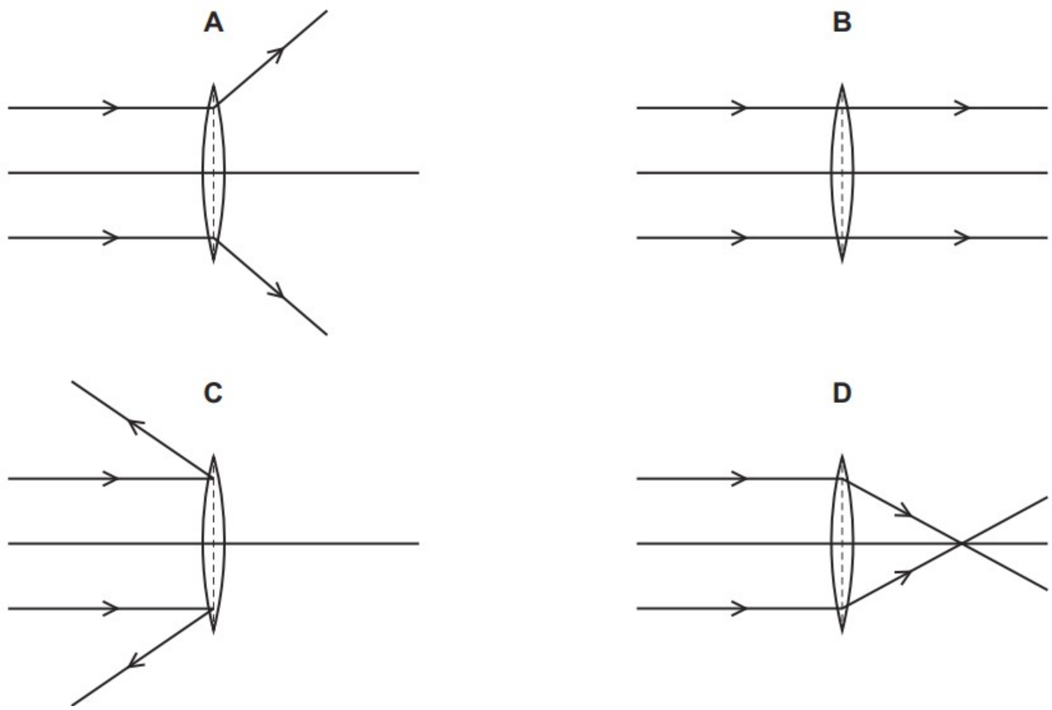
A ray of light is incident on the water surface.

What is the correct path for the ray of light?



44. M/J/20/11/Q28

Which diagram shows the action of a converging lens on a parallel beam of light?



45. M/J/20/11/Q29

What is a feature of red light compared with that of violet light?

A A prism deviates red light more.

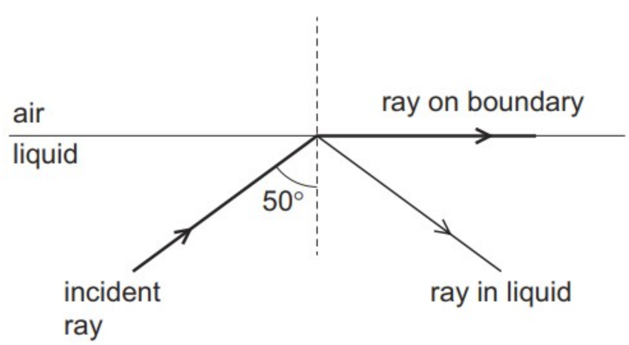
B Red light has a lower frequency.

C Red light has a shorter wavelength.

D The speed of red light in a vacuum is smaller.

46. O/N/19/11/Q26

The diagram shows a ray of light in liquid incident on the boundary with air. Two other rays are observed. One is in the liquid and the other is in the air on the boundary.

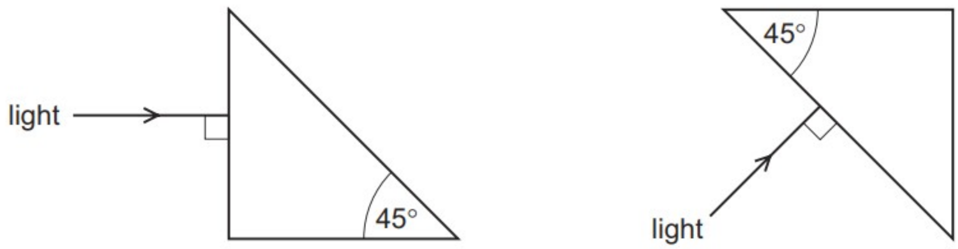


What is the angle of refraction?

A 0° **B** 40° **C** 50° **D** 90°

47. O/N/19/11/Q27

Light is incident at 90° on the surfaces of two glass prisms P and Q.



glass prism P glass prism Q

The critical angle for light travelling from glass into air is 42°.

Where does total internal reflection occur?

A in P and in Q

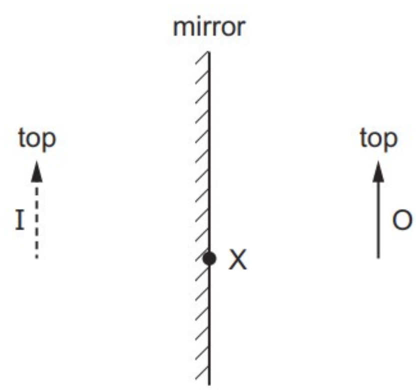
B in P only

C in Q only

D in neither P nor Q

48. O/N/19/12/Q24

An object O is placed in front of a plane mirror. I is the image formed.



A ray from the top of the object is incident on the mirror at X.

What happens to this ray?

A It reflects and passes through the bottom of O.

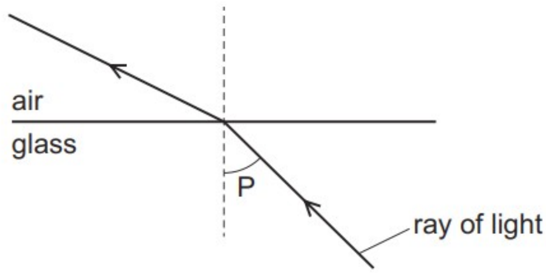
B It reflects and passes through the top of O.

C It reflects as though it came from the bottom of I.

D It reflects as though it came from the top of I.

49. O/N/19/12/Q25

The diagram shows light passing from glass into air.



What is the name of angle P?

A the angle of incidence

B the angle of reflection

C the angle of refraction

D the critical angle

50. M/J/19/12/Q27

An object is placed at a distance from a converging lens that is equal to twice the focal length of the lens.

Which statement about the image is correct?

A It is enlarged.

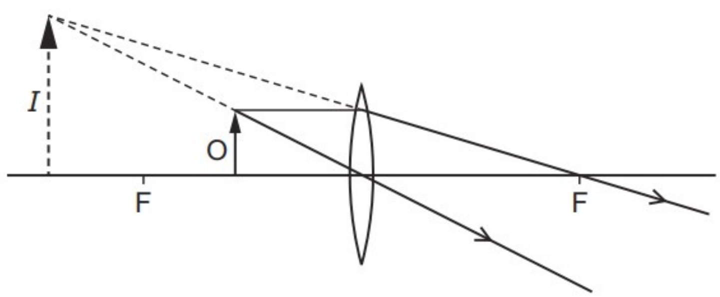
B It is inverted.

C It is on the same side of the lens as the object.

D It is virtual.

51. M/J/19/12/Q28

The lens in the diagram produces an image I of the object O.



Why is this **not** the ray diagram for a photographic enlarger?

A The image is magnified.

B The image is virtual.

C The lens is a converging lens.

D The lens is too thin.

52. M/J/19/11/Q24

A ray of light strikes a plane mirror at an angle of incidence of 20°.

The angle of incidence is then increased by 5°.

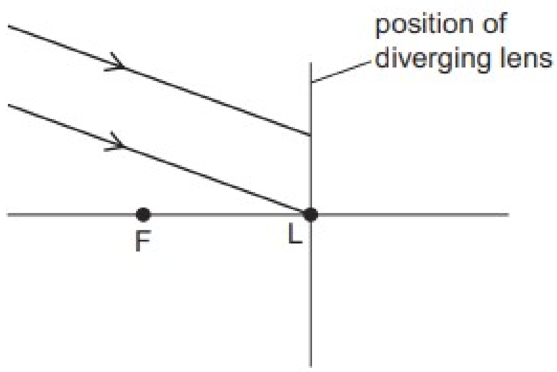
What is the new angle between the incident ray and the reflected ray?

A 10° **B** 25° **C** 45° **D** 50°

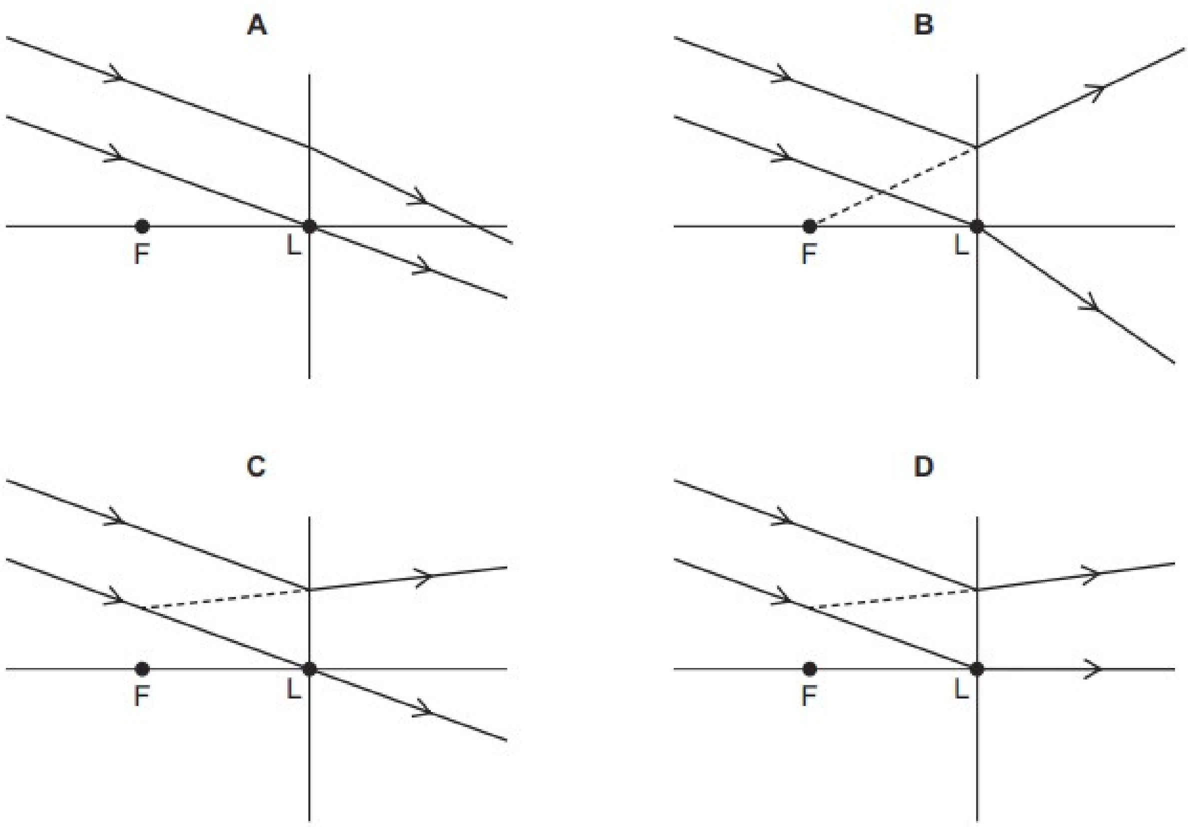
53. M/J/19/11/Q25

A parallel beam of light is incident on a thin diverging lens.

The focal length of the lens is FL, as shown in the diagram.



Which ray diagram shows the beam after it has passed through the lens?



54. M/J/19/11/Q26

The following lists show colours of the spectrum.

Which list shows these colours in order of increasing frequency?

A blue, violet, red, orange, yellow, green

B green, blue, violet, red, orange, yellow

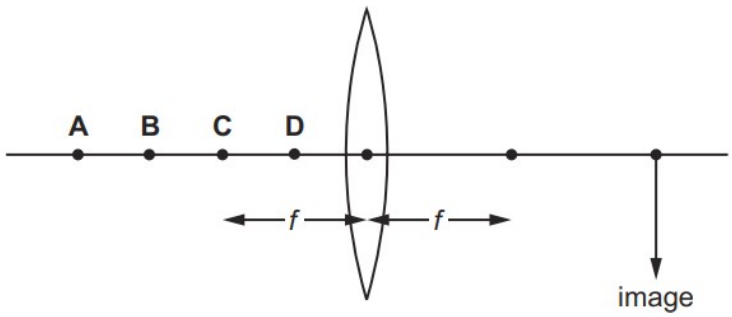
C red, orange, violet, yellow, green, blue

D red, orange, yellow, green, blue, violet

55. O/N/18/11/Q19

The diagram shows a thin converging lens of focal length *f*.

Where must an object be placed to produce a real image in the position shown?



56. O/N/18/11/Q20

White light is dispersed by a prism. Compared with blue light, the **red** light is

A slowed down less and refracted less.

B slowed down less and refracted more.

C slowed down more and refracted less.

D slowed down more and refracted more.

57. ON/18/12/Q25

A student reads the following in her physics book.

> 'The incident angle is greater than 42° which is the critical angle for glass in air.'

What is the student reading about?

A focal length of a glass lens

B reflection in a plane mirror

C refraction as light enters glass

D total internal reflection

58. M/J/18/12/Q28

Which statement about blue light is correct?

A Blue light has a smaller frequency than red light.

B Blue light has a longer wavelength than red light.

C Blue light has the same speed in glass as red light.

D Blue light is refracted more by a glass prism than red light.

59. M/J/18/12/Q29

Which device uses total internal reflection?

A magnifying glass

B optical fibre

C photographic enlarger

D projector

60. M/J/18/11/Q23

Which statement is correct?

A Total internal reflection only occurs when light travels from air into glass.

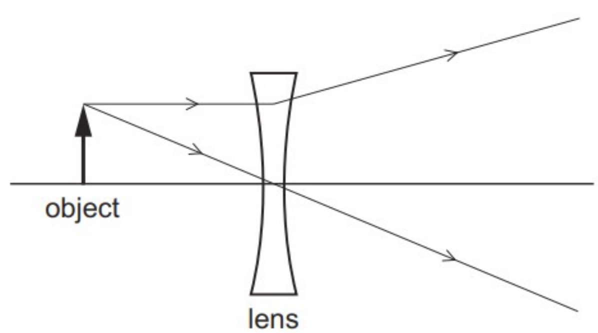
B The larger the refractive index of glass, the larger is the critical angle.

C When total internal reflection occurs, the angle of incidence is equal to the angle of reflection.

D When total internal reflection occurs, the angle of incidence is less than the critical angle.

61. M/J/18/11/Q24

The diagram shows the paths of two rays from the top of an object through a lens.



The object is viewed from the opposite side of the lens to the object.

How does the image compare with the object?

A larger and inverted

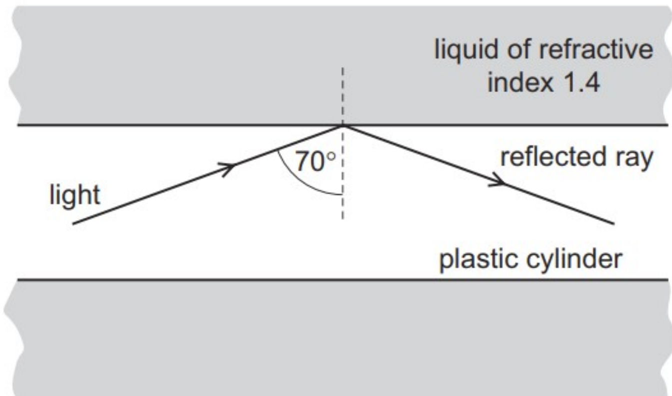
B larger and the same way up

C smaller and inverted

D smaller and the same way up

62. ON/17/11/Q20

A solid plastic cylinder is immersed in a liquid of refractive index 1.4. Light travelling in the plastic cylinder strikes the inside surface at an angle of incidence of 70°. The light undergoes total internal reflection.



What are the values of the critical angle in the plastic and the refractive index of the plastic?

	critical angle in plastic	refractive index of plastic
A	greater than 70°	greater than 1.4
B	greater than 70°	less than 1.4
C	less than 70°	greater than 1.4
D	less than 70°	less than 1.4

63. O/N/17/11/Q21

What is the name and shape of the lens used to correct short sight?

	name of lens	shape of lens
A	converging	
B	converging	
C	diverging	
D	diverging	

64. O/N/17/11/Q22

An object is placed in front of a converging lens. The lens forms a magnified image of the object on a screen.

Which statement is correct?

A The distance between the object and the lens is greater than the focal length.

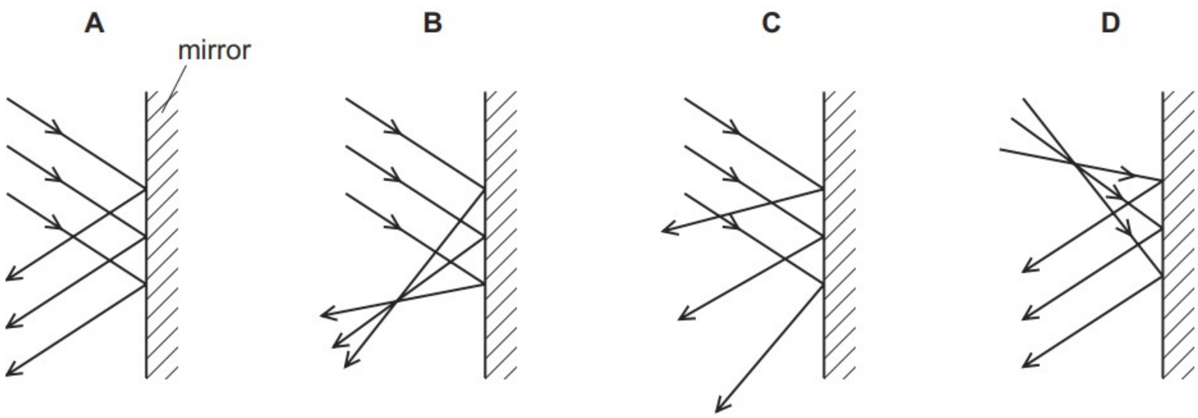
B The image formed is a virtual image.

C The image is the right way up.

D The lens is acting as a magnifying glass.

65. O/N/17/12/Q20

Which diagram shows reflection by a plane mirror?



66. O/N/17/12/Q21

Light is incident on a plastic block of refractive index 1.5.

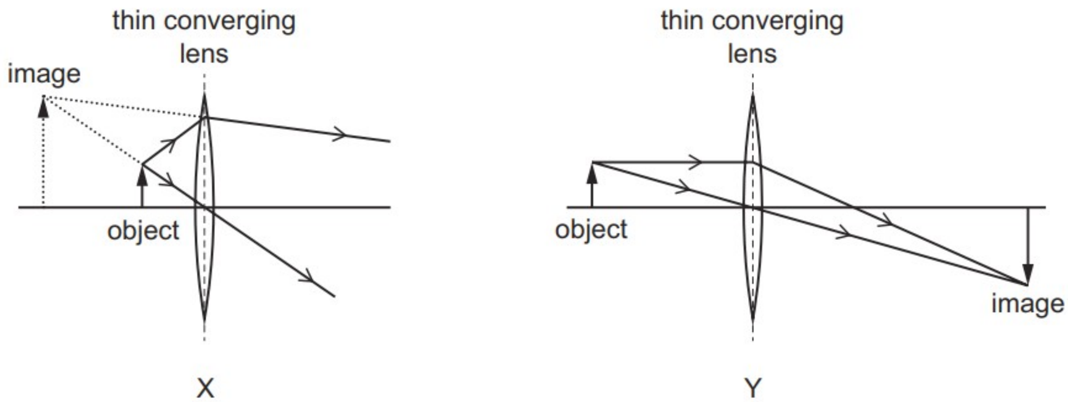
The angle of incidence is 50°.

What is the angle of refraction?

A 31° **B** 33° **C** 40° **D** 75°

67. O/N/17/12/Q23

The ray diagrams, X and Y, show two ways in which a thin converging lens produces an image that is larger than the object.



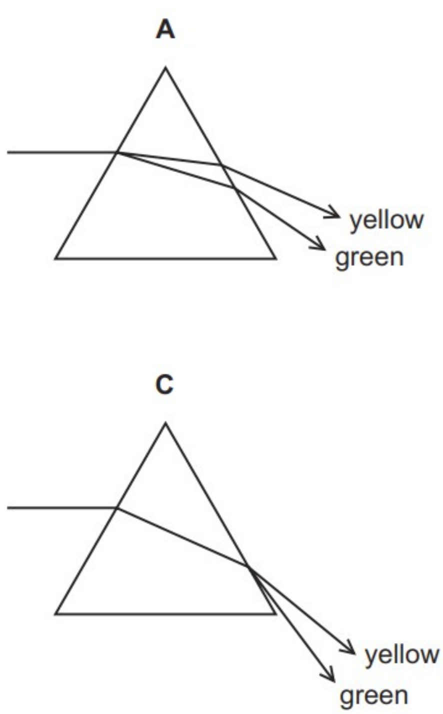
Which devices use a lens as shown in diagram X and in diagram Y?

	X	Y
A	camera	magnifying glass
B	magnifying glass	projector
C	photographic enlarger	camera
D	photographic enlarger	projector

68. M/J/17/12/Q28

A narrow beam of yellow and green light is separated as it passes through a prism.

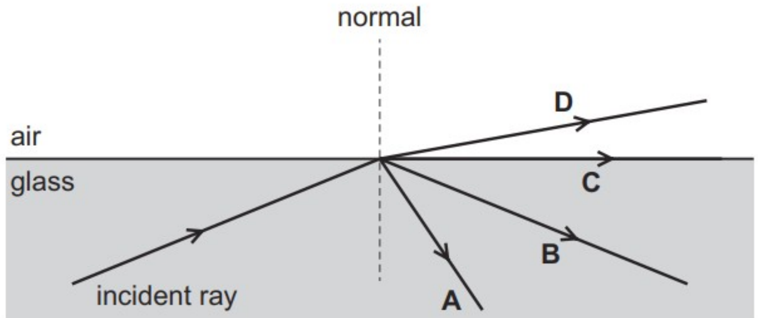
Which ray diagram is correct?



69. M/J/17/11/Q27

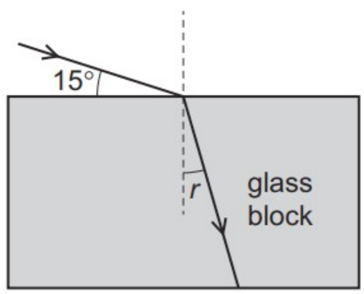
Light travelling in glass is incident on a glass-air boundary. The angle of incidence of the light is greater than the critical angle.

Which arrow shows the direction of the light after it is incident on the boundary?



70. M/J/17/11/Q28

Light strikes the top surface of a glass block at an angle of 15° as shown.



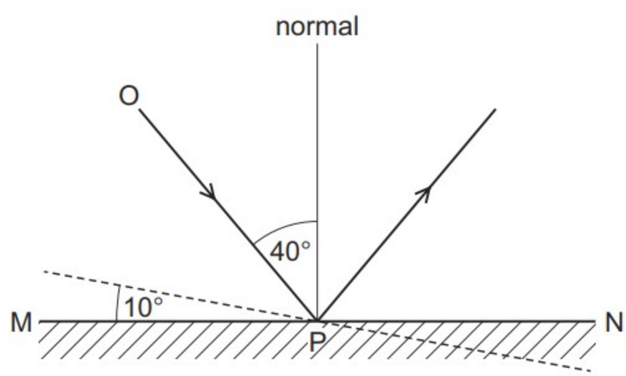
The refractive index of glass is 1.5.

What is the angle of refraction *r*?

A 10° **B** 23° **C** 40° **D** 50°

71. O/N/16/11/Q23

The angle of incidence of ray OP on the plane mirror MN is 40°.



The mirror is rotated through 10°, as shown by the dashed line. The direction of the incident ray OP does not change.

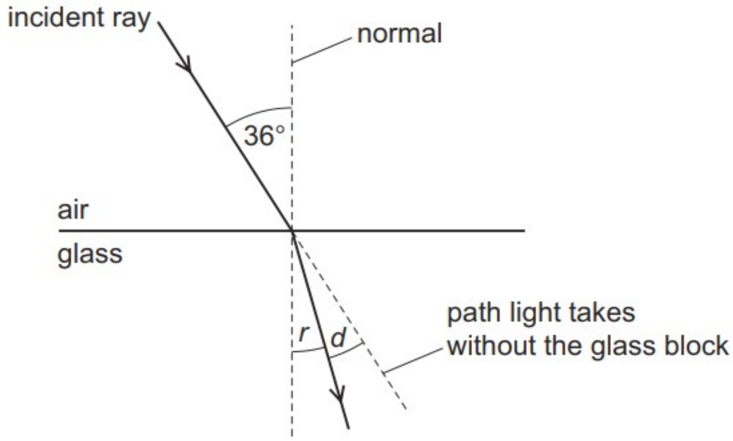
What is the new angle of incidence?

A 30° **B** 40° **C** 50° **D** 60°

72. O/N/16/11/Q24

A ray of light is incident on the surface of a glass block. The diagram is not drawn to scale.

The refractive index of the glass is 1.5.



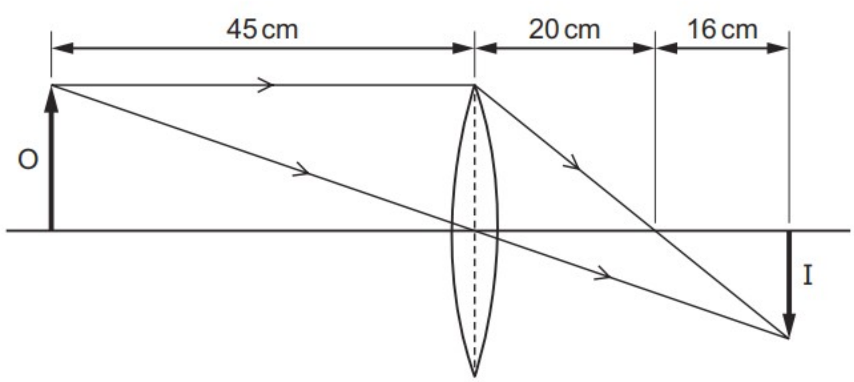
The angle of refraction is *r*. The angle between the refracted ray and the path the light takes without the glass block is *d*.

What are *r* and *d*?

	r / °	d / °
A	23	12
B	24	12
C	23	13
D	24	13

73. O/N/16/11/Q25

In the diagram, a convex lens forms an image I of an object O. The diagram is not drawn to scale.

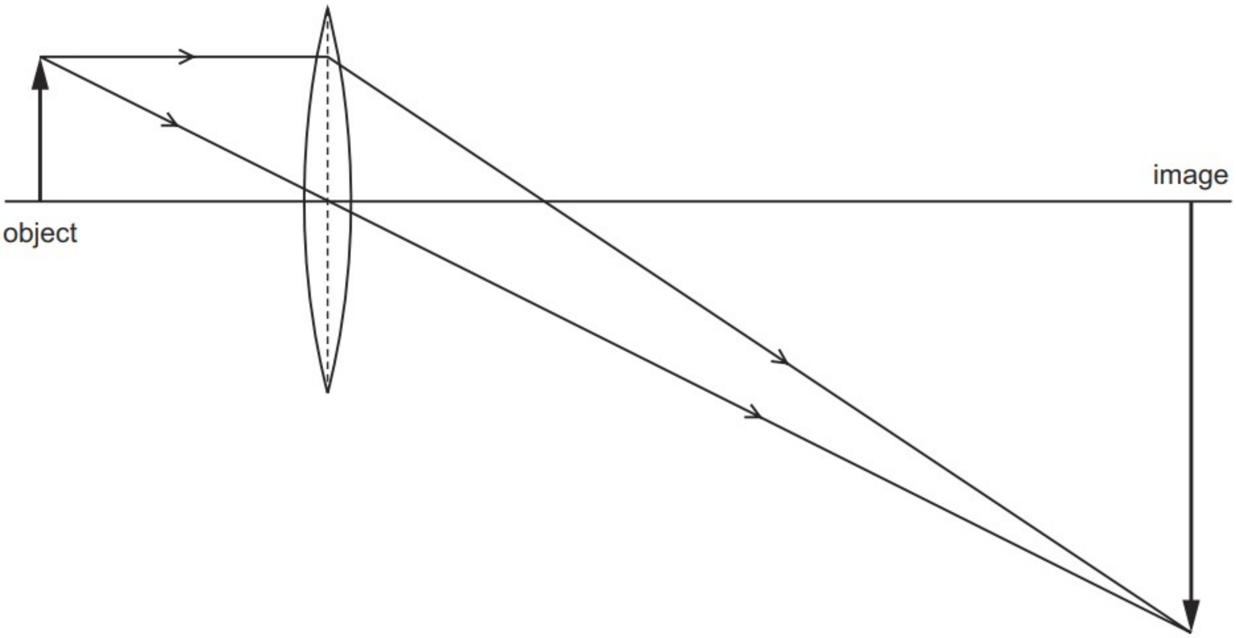


What is the focal length of the lens?

A 16 cm **B** 20 cm **C** 36 cm **D** 45 cm

74. O/N/16/11/Q26

A lens is used to produce a magnified image, as shown in the scale diagram.

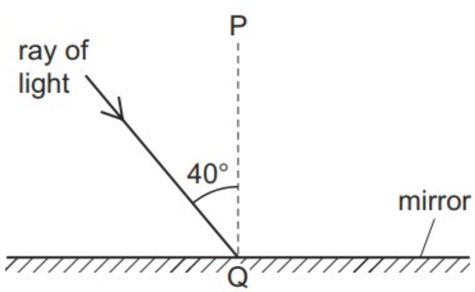


What is the linear magnification produced by the lens?

A 0.33 **B** 3.0 **C** 4.0 **D** 6.0

75. M/J/16/12/Q18

The diagram shows light incident on a plane mirror.

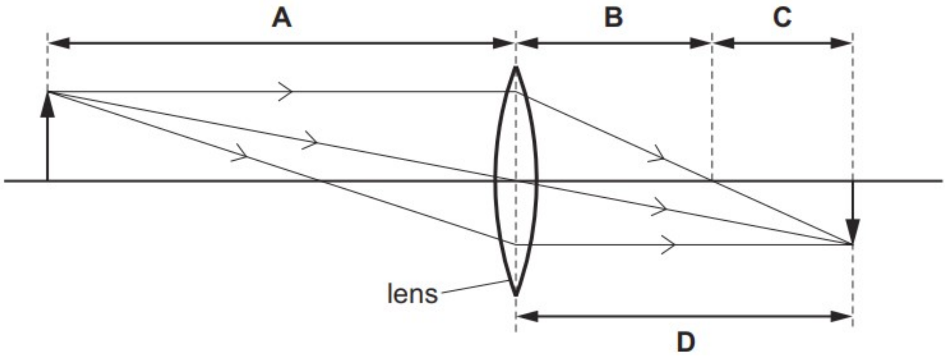


Which row gives the angle of reflection and the name of line PQ?

	angle of reflection	the line PQ is called the
A	40°	normal
B	40°	reflected ray
C	50°	normal
D	50°	reflected ray

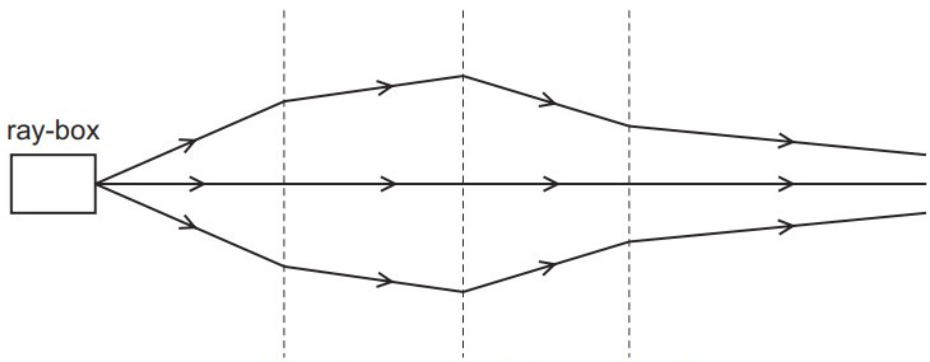
76. M/J/16/12/Q19

Which length is the focal length of the lens shown in the diagram?



77. M/J/16/12/Q20

The rays of light from a ray-box pass through three lenses placed at positions 1, 2 and 3.

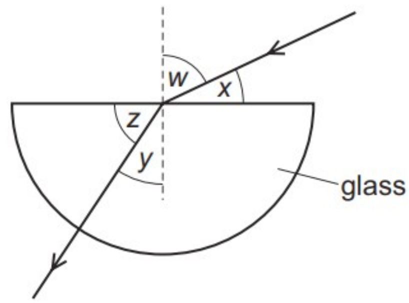


What type of lens is used at each position?

	position 1	position 2	position 3
A	converging	converging	converging
B	converging	converging	diverging
C	diverging	converging	diverging
D	diverging	diverging	converging

78. M/J/16/12/Q21

Light passes from air into a block of glass, as shown.



Which expression is equal to the refractive index of glass?

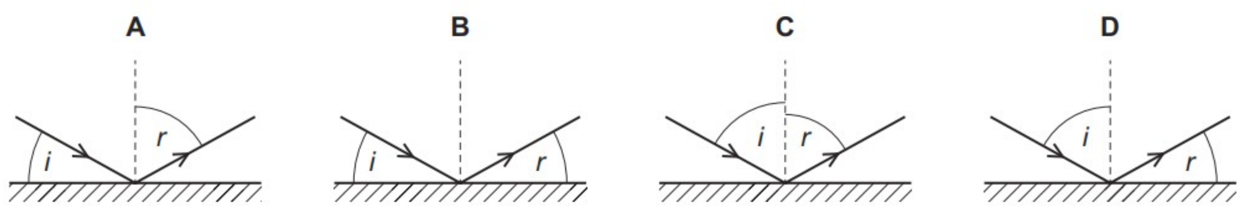
A $\frac{\sin w}{\sin y}$ **B** $\frac{\sin w}{\sin z}$ **C** $\frac{\sin y}{\sin w}$ **D** $\frac{\sin z}{\sin x}$

79. M/J/16/11/Q21

Light is incident on a mirror. The light is reflected from the mirror.

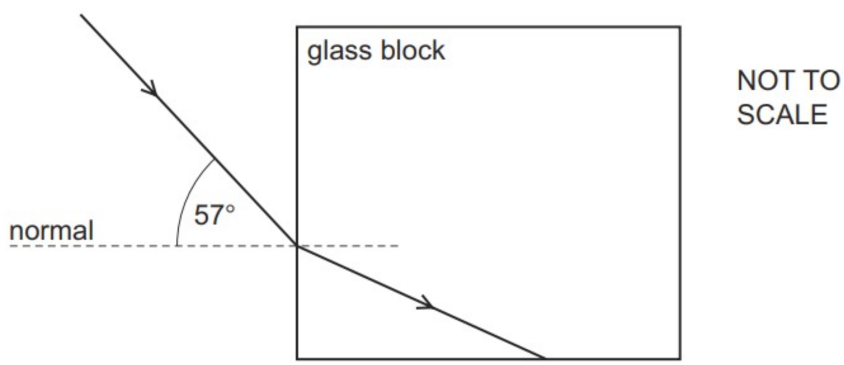
The angle of incidence is i and the angle of reflection is r.

Which diagram correctly shows i and r?



80. M/J/16/11/Q23

Light passes from air into a glass block of refractive index 1.5, as shown.



What is the angle of refraction in the glass and what is the critical angle?

	angle of refraction	critical angle
A	34°	42°
B	34°	60°
C	38°	42°
D	38°	60°

81. O/N/15/11/Q20

Which statement about red light and blue light is correct?

A Red light has a higher frequency than blue light.

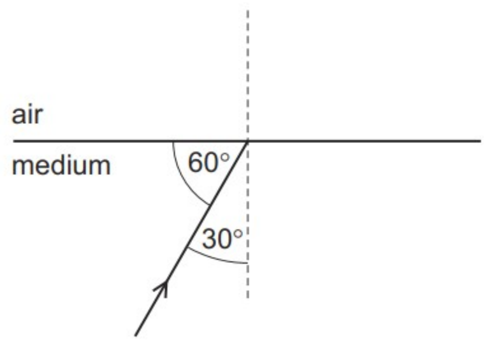
B Red light has a longer wavelength than blue light.

C Red light has the same speed in glass as blue light.

D Red light is refracted by a glass prism more than blue light.

82. O/N/15/11/Q21

A ray of light in a transparent medium of refractive index 1.8 is incident on the surface as shown. The light enters air.



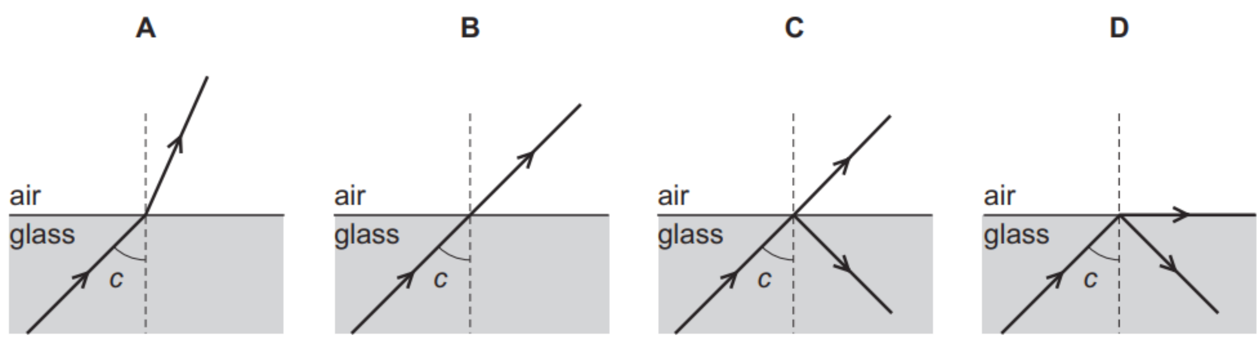
What is the angle between the refracted ray and the normal in air?

A 29° **B** 33° **C** 54° **D** 64°

83. M/J/15/12/Q25

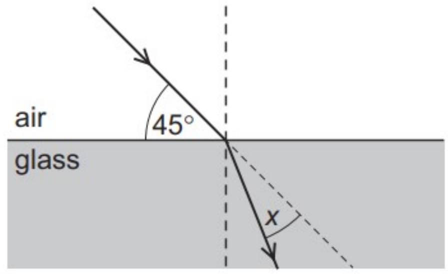
A ray of light in glass is incident on the surface at an angle *c*. The angle *c* is the critical angle.

Which diagram shows what happens to the light?



84. M/J/15/12/Q26

A ray of light is incident on the surface of a glass block, as shown in the diagram below.



The refractive index of the glass is 1.5.

The light ray changes direction when entering the glass.

What is the angle *x* through which the ray moves?

A 30° **B** 28° **C** 17° **D** 15°

85. M/J/15/12/Q27

An image is formed by a thin converging lens when it is used as a magnifying glass.

What is the correct description of the image?

A real and erect

B real and inverted

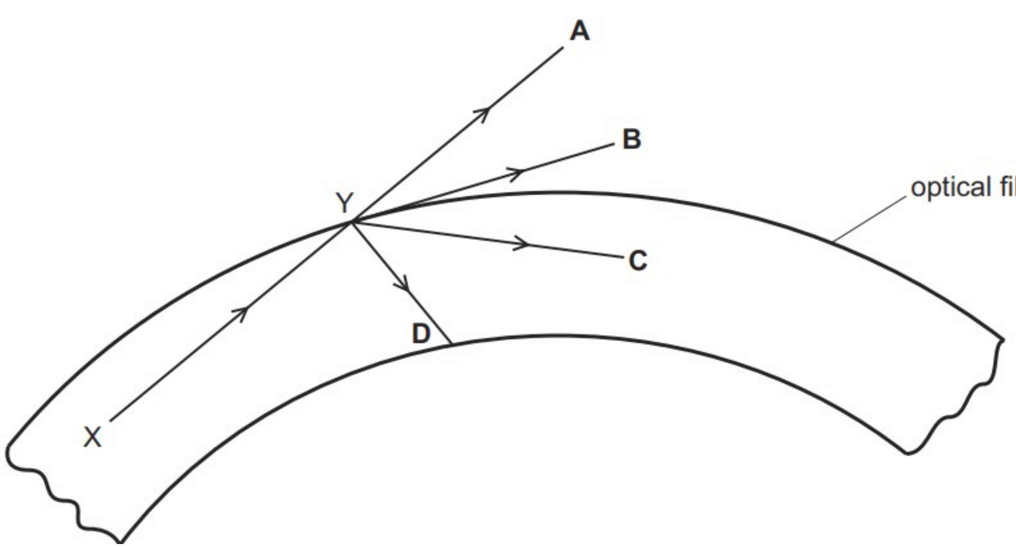
C virtual and erect

D virtual and inverted

86. M/J/15/11/Q24

A ray of light travels from X to Y along an optical fibre. The angle of incidence at Y is greater than the critical angle.

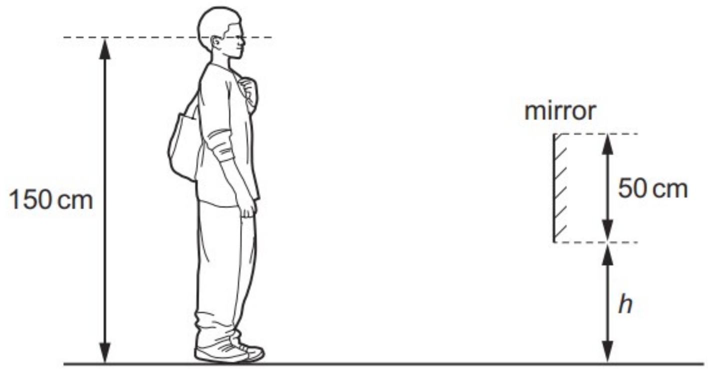
In which direction does the ray of light travel after reaching point Y?



87. O/N/14/11/Q20

A shoe shop puts a mirror on the wall so that customers can look at their shoes.

The length of the mirror is 50 cm. A customer has eyes 150 cm above ground level.



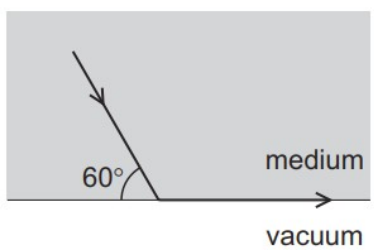
The bottom of the mirror is at height *h* above the ground.

What is the smallest value of *h* that allows the customer to see an image of his shoes in the mirror?

A 0 **B** 25 cm **C** 50 cm **D** 75 cm

88. O/N/14/11/Q21

The diagram shows light travelling through a medium. The light reaches the boundary with a vacuum as shown. The light emerges travelling along the surface.



What is the refractive index of the medium?

A $\dfrac{\sin 60°}{\sin 30°}$ **B** $\dfrac{\sin 60°}{\sin 90°}$ **C** $\dfrac{\sin 90°}{\sin 30°}$ **D** $\dfrac{\sin 90°}{\sin 60°}$

89. O/N/14/11/Q23

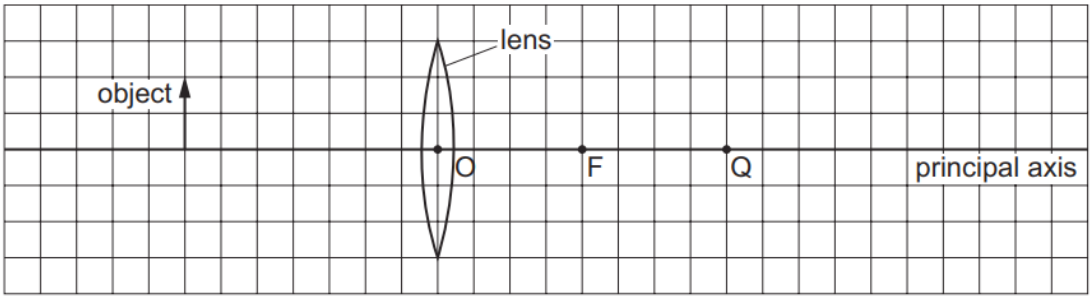
A digital camera uses a lens to produce a diminished (reduced in size) image on a light sensor.

Which row shows the correct type of lens and the nature of the image?

	type of lens	nature of image
A	converging	inverted
B	converging	upright
C	diverging	inverted
D	diverging	upright

90. O/N/14/11/Q22

The diagram shows an object on the principal axis of a converging (convex) lens. A principal focus of the lens is at F.



Where is the image formed by the lens?

A between O and F

B between F and Q

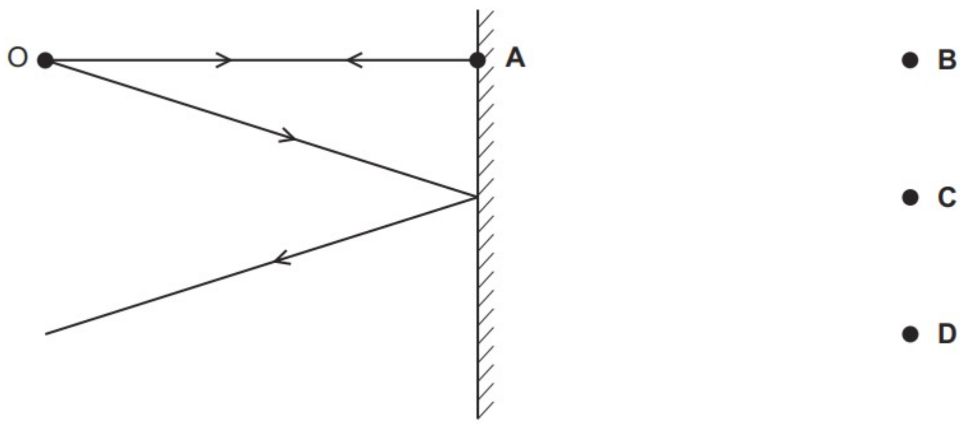
C at Q

D to the right of Q

91. M/J/14/12/Q26

The diagram shows two divergent rays of light from an object O being reflected from a plane mirror.

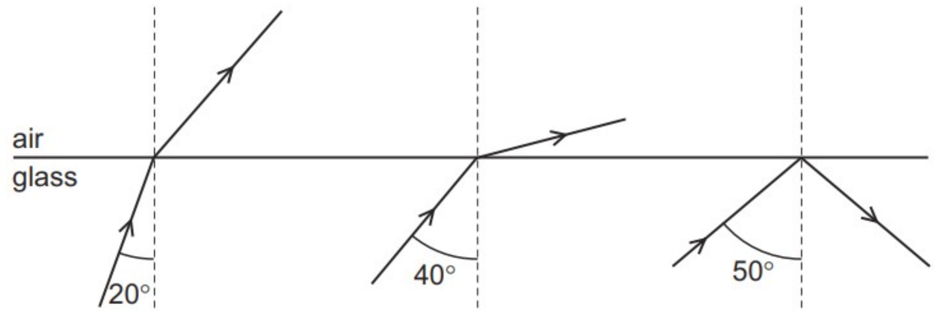
At which position is the image formed?



92. M/J/14/11/Q25

Three rays of light are incident on the boundary between a glass block and air.

The angles of incidence are different.



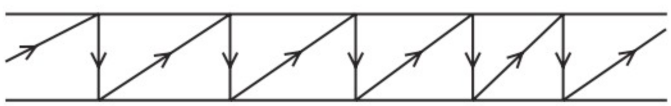
What is a possible critical angle for light in the glass?

A 15° **B** 30° **C** 45° **D** 60°

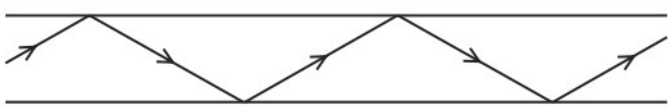
93. O/N/13/11/Q20

Which diagram represents the reflection of light along an optical fibre?

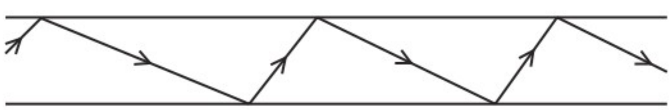
A



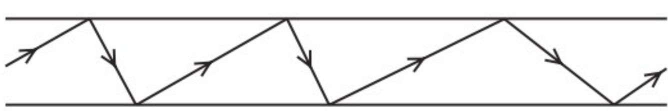
B



C

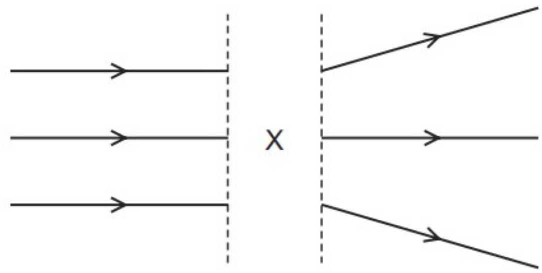


D



94. O/N/13/11/Q21

The diagram shows rays of light.



What is in the space labelled X?

A a converging lens

B a diverging lens

C a plane mirror

D a rectangular glass block

95. O/N/13/12/Q24

A ray of light strikes the surface of a glass block at an angle of incidence of 45°.

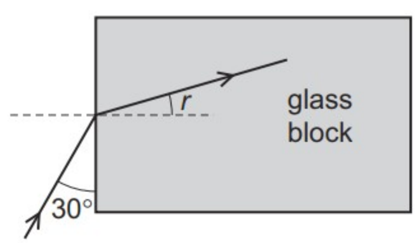
The refractive index of the glass is 1.8.

What is the angle of refraction inside the block?

A 23° **B** 25° **C** 45° **D** 81°

96. M/J/13/11/Q20

A ray of light meets the face of a glass block at an angle of 30° as shown.



The refractive index of the glass is 1.5.

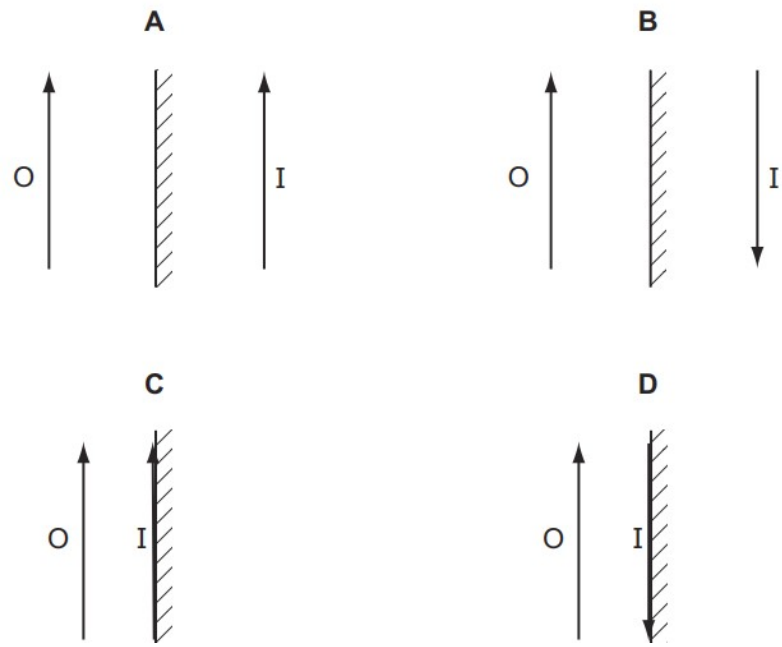
What is the angle of refraction *r* inside the glass block?

A 19° **B** 20° **C** 35° **D** 40°

97. O/N/12/11/Q22

An object O is placed in front of a plane mirror.

Which diagram correctly represents the image I formed by the mirror?



98. O/N/12/11/Q25

Red and violet are the colours at the ends of the visible spectrum.

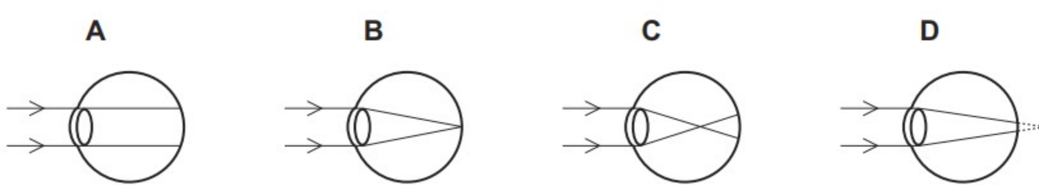
How do the frequencies and the wavelengths of these colours compare?

	higher frequency	longer wavelength
A	red	red
B	red	violet
C	violet	red
D	violet	violet

99. M/J/12/12/Q24

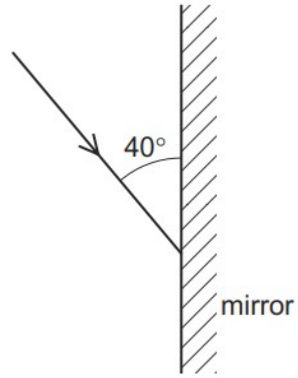
A man is short-sighted.

Which ray diagram shows what happens in his eye when he looks at a distant object?



100. M/J/12/12/Q22

The diagram shows a ray of light directed at a plane mirror.



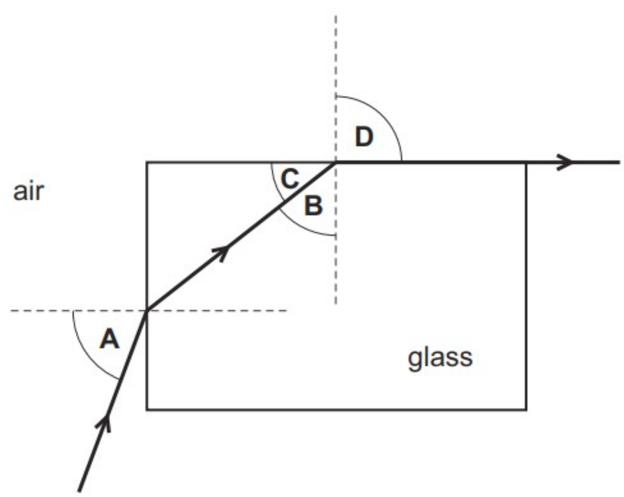
What are the angle of incidence and the angle of reflection?

	angle of incidence	angle of reflection
A	40°	40°
B	40°	50°
C	50°	40°
D	50°	50°

101. M/J/12/12/Q23

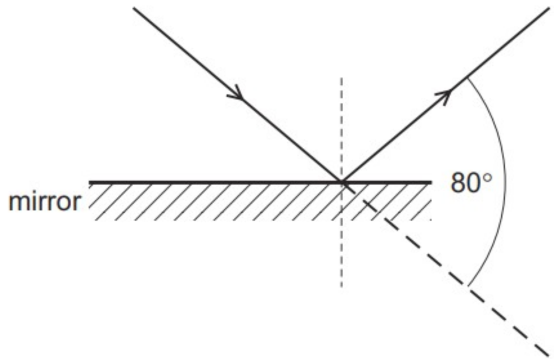
Light travels through a glass block as shown.

Which angle is the critical angle for light in the glass?



102. M/J/12/11/Q22

Light is incident on a mirror and is reflected as shown.



What is the angle of incidence and the angle of reflection?

	angle of incidence / °	angle of reflection / °
A	40	40
B	40	50
C	50	40
D	50	50

103. M/J/12/11/Q23

Light is incident on one face of a glass block at an angle of incidence of 40°. The glass block is in air.

The refractive index of the glass is 1.46.

What is the angle of refraction inside the glass block?

A 26° **B** 27° **C** 58° **D** 70°

104. M/J/12/11/Q24

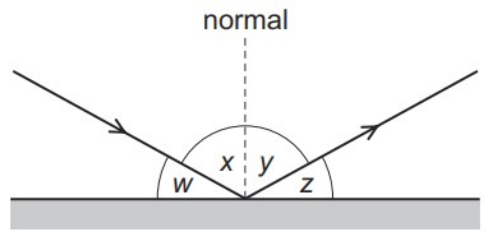
In a short-sighted eye, light from distant objects is not focused on the retina.

Where is this light focused and what type of lens is needed to correct the problem?

	where focused	lens needed
A	behind the retina	converging lens
B	behind the retina	diverging lens
C	in front of the retina	converging lens
D	in front of the retina	diverging lens

105. M/J/11/12/Q23

A ray of light strikes a plane mirror and is reflected.

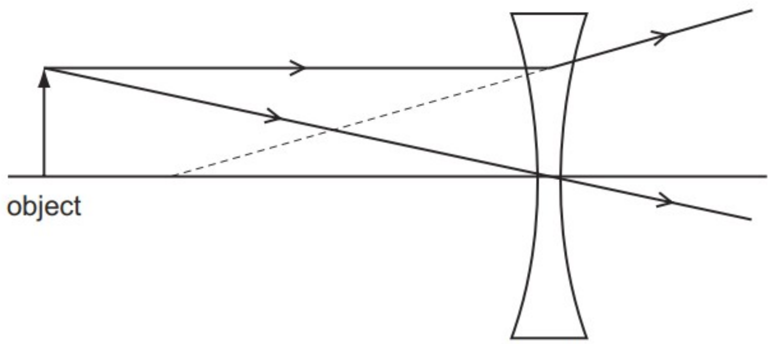


Which pair of angles **must** be equal in value?

A *w* and *x* **B** *w* and *y* **C** *x* and *y* **D** *x* and *z*

106. M/J/11/12/Q24

The ray diagram shows two rays from a point on an object placed in front of a diverging (concave) lens.



What are the properties of the image produced?

A real and larger than the object

B real and smaller than the object

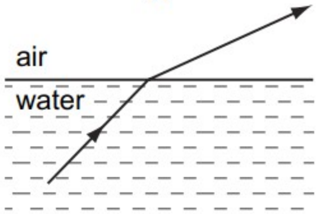
C virtual and larger than the object

D virtual and smaller than the object

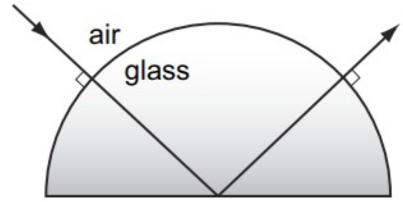
107. M/J/11/11/Q22

In which diagram is the path of the light ray **not** correct?

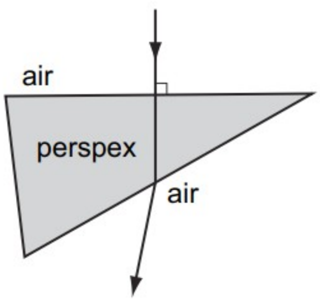
A



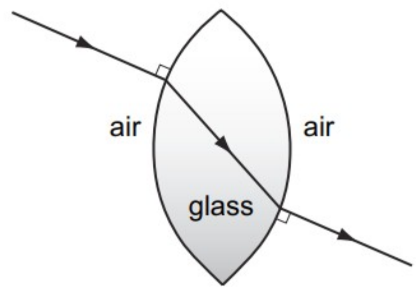
B



C



D



108. O/N/10/11/Q22

A ray of light strikes the surface of a glass block at an angle of incidence of 45°.

The refractive index of the glass is 1.5.

What is the angle of refraction inside the block?

A 28° **B** 30° **C** 45° **D** 67°

109. O/N/10/11/Q23

An object is viewed through a concave (diverging) lens.

What is the correct description of the image formed?

A real, inverted, magnified

B real, upright, diminished

C virtual, inverted, magnified

D virtual, upright, diminished

110. M/J/10/12/Q22

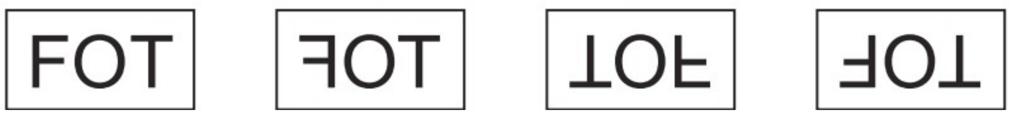
A student holds a sheet of paper with letters on it facing a plane mirror.

The letters on the paper are shown.

TOF

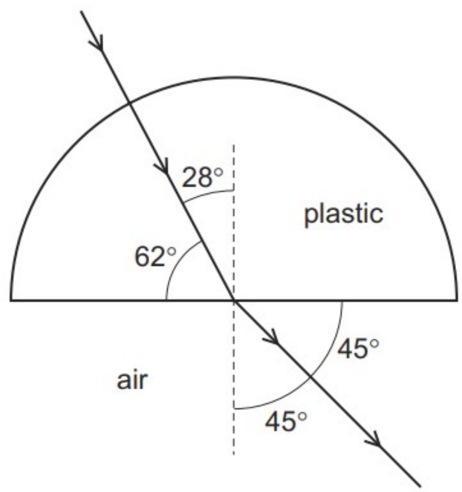
What does the student see in the mirror?

A B C D



111. M/J/10/12/Q23

A semi-circular block is made from a plastic. A ray of light passes through it at the angles shown.

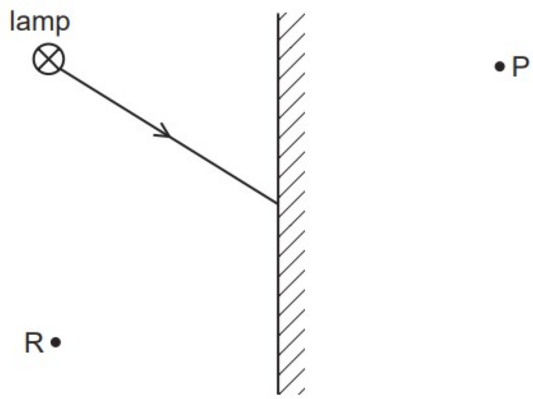


To two decimal places, what is the refractive index of the plastic?

A 1.25 **B** 1.41 **C** 1.51 **D** 1.61

112. O/N/09/Q20

The diagram shows a ray of light from one point on a lamp striking a plane mirror.



The image of the point on the lamp formed by the mirror is

A at P and is real.

B at P and is virtual.

C at R and is real.

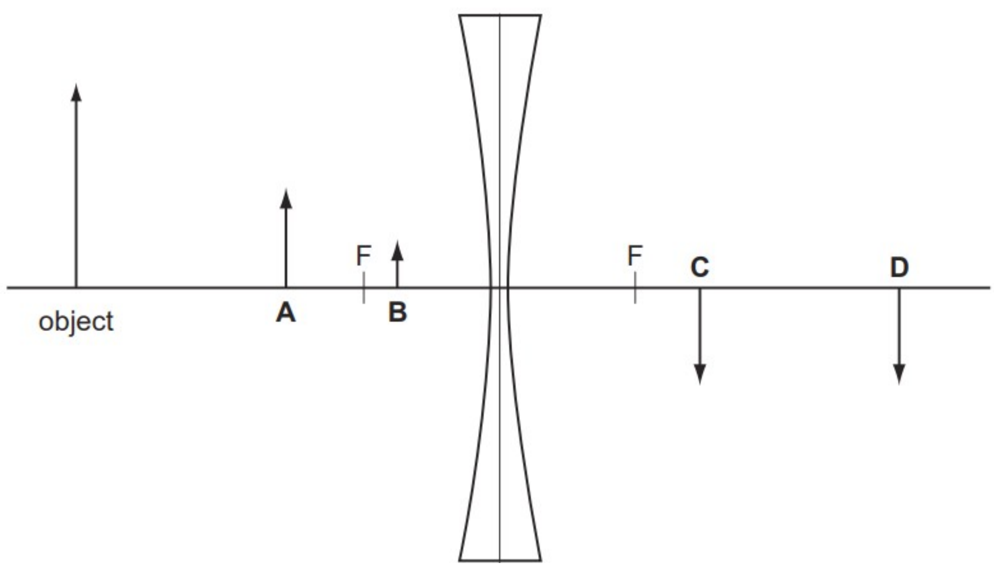
D at R and is virtual.

113. O/N/09/Q22

An object is placed in front of a diverging lens as shown on the scale diagram.

The principal focus F is marked on each side of the lens.

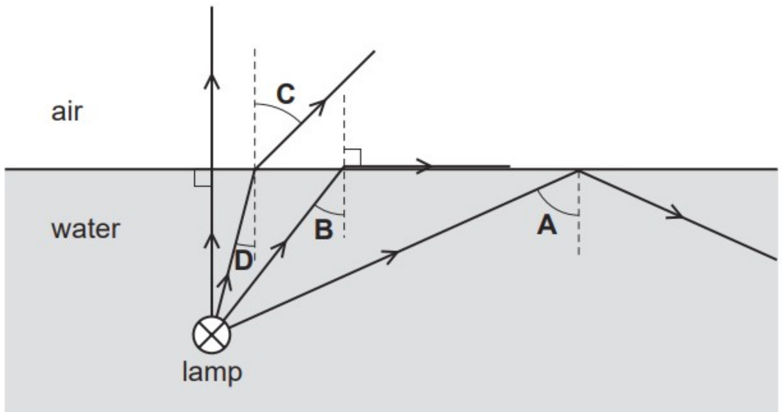
What is the position of the image formed by the lens?



114. M/J/09/Q21

The diagram shows four rays of light from a lamp below the surface of some water.

What is the critical angle for light in water?



115. M/J/09/Q22

When white light is dispersed by a prism, compared with blue light, the **red** light is

A slowed down less and refracted less.

B slowed down less and refracted more.

C slowed down more and refracted less.

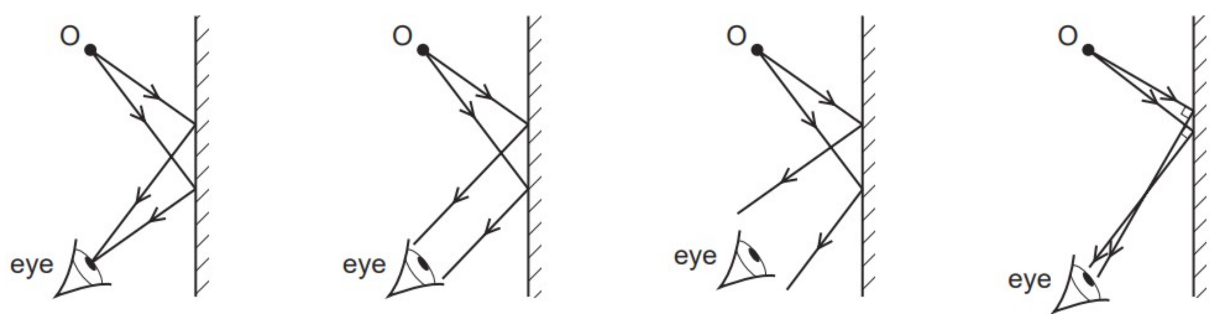
D slowed down more and refracted more.

116. O/N/08/Q23

An eye views an object O by reflection in a plane mirror.

Which is the correct ray diagram?

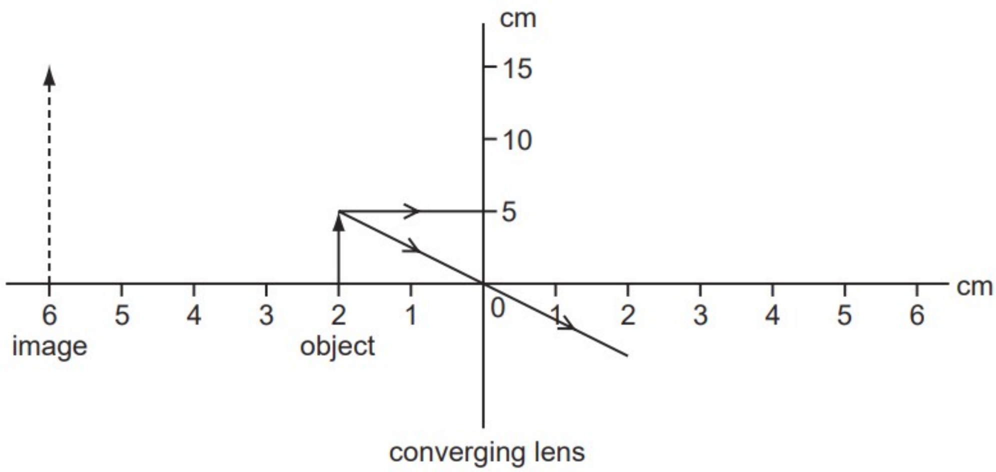
A B C D



117. M/J/08/Q23

An object 5.0 cm high is placed 2.0 cm from a converging (convex) lens which is being used as a magnifying glass.

The image produced is 6.0 cm from the lens and is 15 cm high.



What is the focal length of the lens?

A 2.0 cm **B** 3.0 cm **C** 4.0 cm **D** 6.0 cm

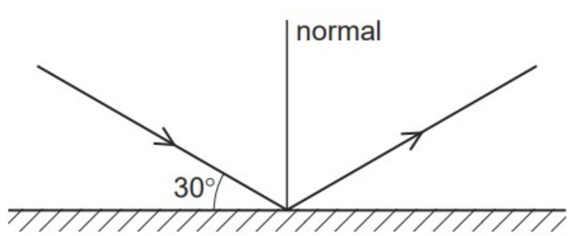
118. M/J/08/Q25

Which colour, red or blue, has the higher frequency and which has the longer wavelength?

	higher frequency	longer wavelength
A	blue	blue
B	blue	red
C	red	blue
D	red	red

119. O/N/07/Q20

The diagram shows a ray of light reflected from a plane mirror.

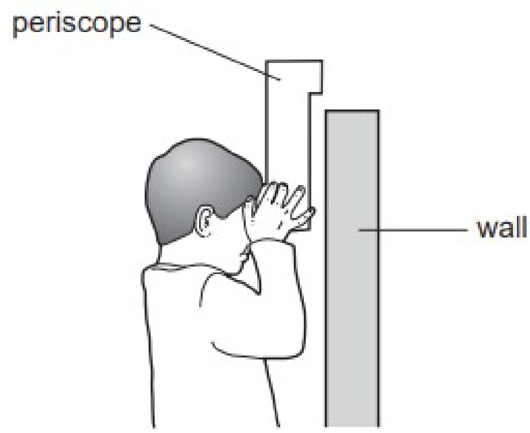


What is the angle of reflection?

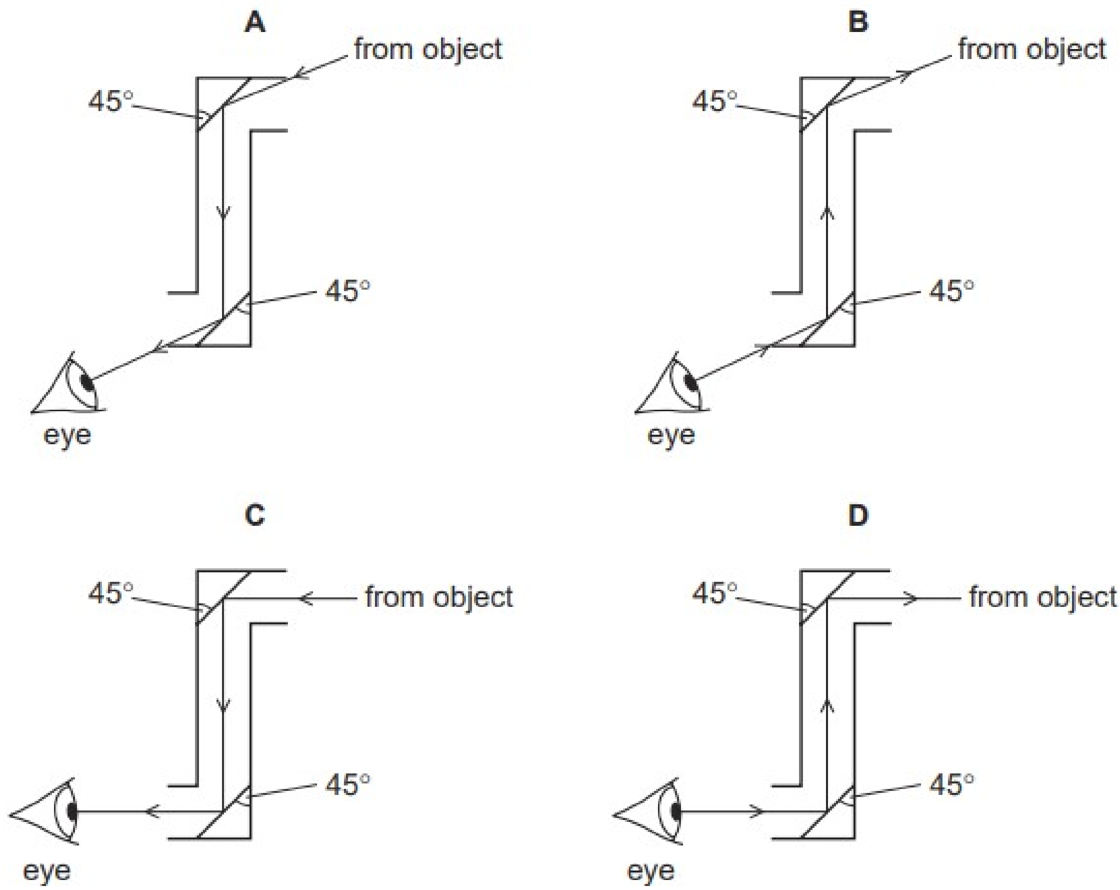
A 30° **B** 60° **C** 90° **D** 120°

120. M/J/07/Q20

The diagram shows a child using a periscope to look at an object on the other side of a wall.



Which diagram shows a correctly drawn ray of light from the object?



121. M/J/07/Q21

What happens to light as it passes from glass into air?

A Its frequency decreases because its speed decreases.

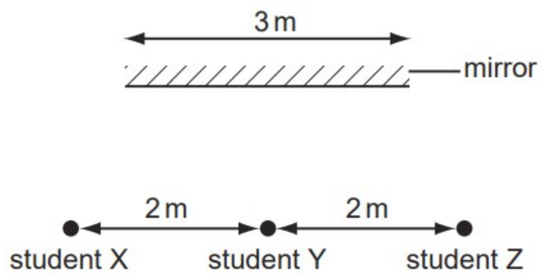
B Its frequency increases because its speed increases.

C Its wavelength decreases because its speed decreases.

D Its wavelength increases because its speed increases.

122. O/N/06/Q21

Three students stand 2 m apart in front of a plane mirror which is 3 m long.



Student Y is standing opposite the mid-point of the mirror.

How many students can see the images of the other two?

A 0 **B** 1 **C** 2 **D** 3

123. O/N/06/Q22

The human eye has a converging lens system that produces an image at the back of the eye.

An eye views a distant object. What type of image is produced?

A real, erect, same size

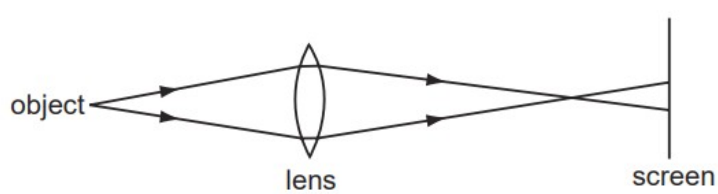
B real, inverted, diminished

C virtual, erect, diminished

D virtual, inverted, magnified

124. M/J/06/Q23

A lens forms a blurred image of an object on a screen.



How can the image be focussed on the screen?

A by moving the object away from the lens and screen

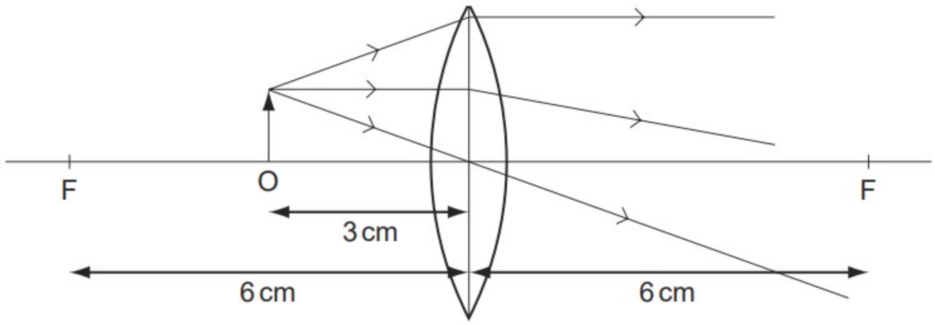
B by moving the screen away from the lens and object

C by using a brighter object at the same position

D by using a lens of longer focal length at the same position

125. O/N/05/Q23

The diagram shows an object O placed 3 cm away from a converging lens of focal length 6 cm.



What type of image is produced?

A real, erect and diminished

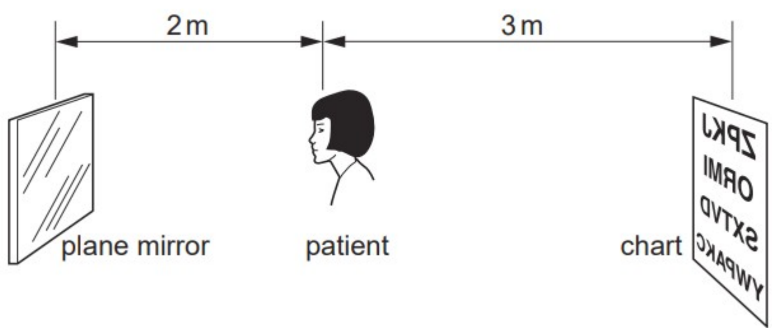
B real, inverted and magnified

C virtual, erect and magnified

D virtual, inverted and diminished

126. M/J/05/Q21

The diagram shows a patient having her eyes tested. A chart with letters on it is placed behind her and she sees the chart reflected in a plane mirror.



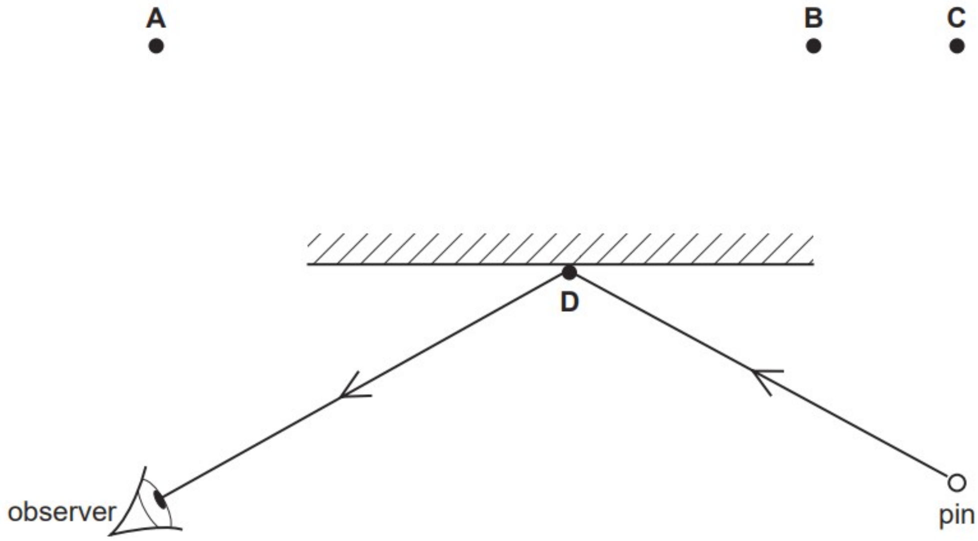
How far away from the patient is the image of the chart?

A 2 m **B** 4 m **C** 5 m **D** 7 m

127. O/N/04/Q24

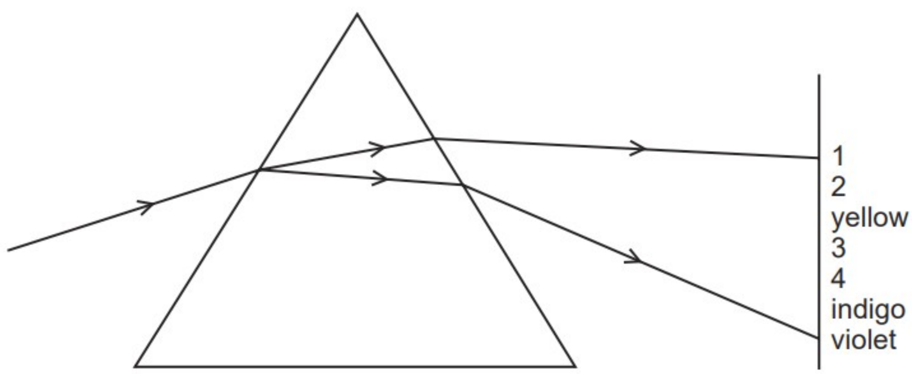
A pin is placed in front of, and to the right of, a plane mirror as shown.

Where is the image of the pin?



128. O/N/04/Q26

The diagram shows the spectrum produced when white light is dispersed by a glass prism.

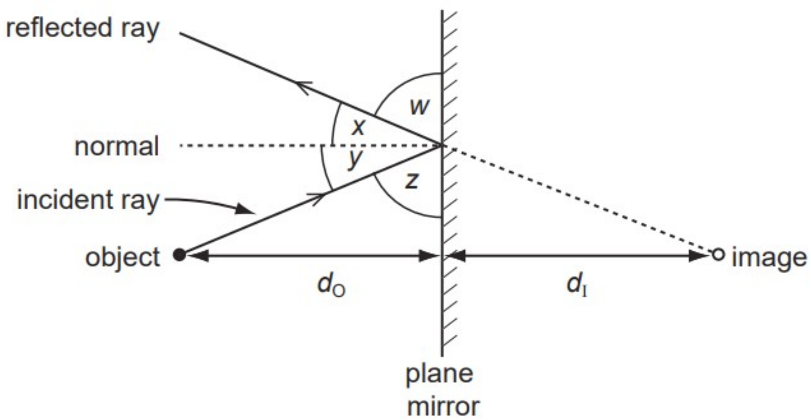


What are the numbered visible colours?

	1	2	3	4
A	infra-red	red	green	ultra-violet
B	red	green	orange	blue
C	red	orange	green	blue
D	red	orange	green	ultra-violet

129. M/J/04/Q20

An image is formed in a plane mirror.

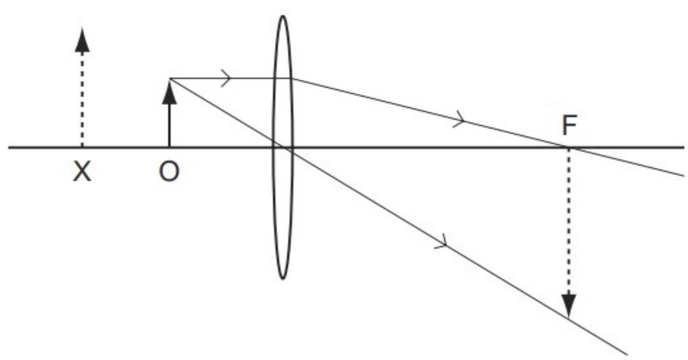


Which statement must be correct?

	angles	distances
A	$w = y$	$d_O = d_I$
B	$w = z$	d_O is greater than d_I
C	$x = y$	$d_O = d_I$
D	$x = z$	d_O is greater than d_I

130. M/J/04/Q22

A student starts to draw a ray diagram for an object at O, near a thin convex lens, but is not sure whether the image is formed at X or at F.



The correctly drawn image is

A real and formed at F.

B real and formed at X.

C virtual and formed at F.

D virtual and formed at X.